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ZMP — Where Are We After Fifty-Five Years?

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Twenty years ago in the first issue of IJHR, among others, there was a published paper [M. Vukobratović and B. Borovac, Zero-moment point — Thirty five years of its life, *Int. J. Humanoid Robot.* **1**(1) (2004) 157–173], devoted to well-known zero-moment point (ZMP) concept. Its role in biped locomotion (in the preservation of dynamic balance during walking and gait synthesis) was (and still is) extremely significant. This paper is aimed to present, how far we have gone in the past 20 years in the realization of bipedal walking, what are remaining open challenges and what we can expect in the further development. At the beginning of the paper, we will briefly refer to some still existing misinterpretations of the ZMP notion, related to specific boundary cases, when the terms Center of Pressure (CoP) and ZMP are still being mixed. Then, we will point your attention to the discussion of two important issues: attempts of generalization of ZMP notion in case of motion in the presence of multiple, spatially distributed and non-planar contacts and walk realization in a non-laboratory environment (unstructured environment and on the non-flat ground surface). Of course, looking at "how humans do it" has always been an inspiration for the realization of complex tasks by humanoids. Learning to walk is a task where several successful realizations have appeared only recently, so we will look at this topic at the end of this paper.

Keywords: Biped motion; unstructured environment; spatially distributed contacts; ZMP.

1. Introduction

Looking back for over half a century we can see how research directions evolved and gain insight into the overall progress of biped's motion — mostly humanoids. In the beginning, the goal was to synthesize gait and perform a dynamically

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balanced walk[a] on a real robot in the lab environment. Today, requirements are much higher and the spectrum of possible applications is much wider. Attempts are being made to make robots coexist in the same living and working space with humans in different environments: in a person's home, in public spaces (department stores, airports, schools, hospitals, parks, etc.), factories, restaurants, etc. Such diversity of applications caused high increase of different research topics. Almost all applications, besides dynamic balance, include a number of different subtasks which have to be solved to enable walking "out of the lab". The basic and most illustrative example is walking in an unstructured environment. Let us clarify this. If in an unstructured environment, the robot should move in a space where it has not been before or a detailed map is unknown to it — in this case, programming the robot's motion in advance is not possible. This means that the robot only knows a desired global direction and a goal target, while the details related to path and motion realization have to be determined online. As soon as the decision about the following movement is made it should be realized, almost immediately. This is just a basic illustration of walking "out of the lab" but the list of problems (tasks) that need to be solved in real time during the walk is much longer. Just illustrative list includes: walking along a route with a large number of subjects (i.e., crowded environment), walking on a surface where obstacles must be avoided or stepped over while walking (e.g., damaged parts of the pavement where there are holes), walking where places, where the foot will be lowered, should be chosen (e.g., when lining up stones or bricks to cross a muddy section), moving in areas where there are technical requirements that must be respected (e.g., coming up on and off escalators, having to wait while automatic doors open, etc.).

Quite a different type of walking-related tasks includes grasping and manipulation during walking. It is often required to involve the hands during motion: from simply holding handrails to make walking easier and safer by facilitating the preservation of dynamic balance, to leaning against the walls in the space where the robot moves because without additional support, movement would not be possible (for example, walking on surfaces with a large slope). A very common task for robots that work in the human environment (we know this from our daily experience) is that the robot needs to pick up an object during motion (which can be heavy, such as a

[a]In recent literature, we can see, more and more frequently, the use of the term "stable walk". This is just a short note about the appropriateness of the term "stable walk" versus "dynamically balanced walk". Control theory gives us several precise definitions of the concepts of stability (e.g., Lyapunov stability or bounded input, bounded output). However, in the case of bipedal walking, the situation is quite different. When we say that the walk is stable, we intuitively mean that the robot is not in danger of falling, and not that the joints follow the predefined trajectories accurately. Even if we have perfect tracking of joint trajectories, the robot can fall. Conversely, in the case of a disturbance, the system can even abandon the realization of the previous motion, e.g., by stepping aside and when the disturbance stops return to the realization of the original motion. Whether the humanoid is in danger of possible falls or not, we can see by the position of the ZMP. The humanoid will not fall (it will stay on its feet) if all the forces acting on the system (including reaction on disturbance) are balanced to ensure that the ZMP is located within the support polygon. This is the reason we believe that the term "dynamically balanced walk" is more appropriate than "stable walk".

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suitcase) and take it to the target location. Walking with such an additional load is quite different than without it, and its realization (preserved dynamic balance is a prerequisite) represents an interesting research task. A quite different task is the joint work of man and robot, e.g., when they jointly carry a large object. This introduces additional restrictions on the movements that can be applied, because contacts of hands and objects during motion should be preserved and the motions of robot and its coworker should be synchronized. In this case, spatially distributed additional contact(s) are above the ground surface and the original zero-moment point (ZMP) notion cannot be employed. Therefore, a different approach is required.

All these research tasks emerged when the concept of ZMP was introduced and the synthesis of bipedal gait was made possible. In our paper[1] 20 years ago we were dealing with the interpretation of ZMP notion, pointing to misunderstandings we noticed and offering a correct interpretation. However, insufficiently precise interpretations still sometimes appear and therefore we provide an illustration of the basic ZMP notion at the beginning of this section. We believe that the task of realizing walking and maintaining dynamic balance has reached its maturity and that further improvements can be expected by solving tasks arising from realizing walking in an unstructured environment or walking in the presence of spatially distributed contacts. We have already mentioned these tasks.

The first topic that we will pay attention to in this paper is motion in the presence of spatial contacts with the environment (including motion planning and contact repositioning), the second one is the realization of motion in an unstructured environment. In addition, we will briefly consider learning-based approaches to robot bipedal walking, which seems very promising.

2. Basics of ZMP Notion — What is ZMP?

ZMP notion,[1] introduced 55 years ago, was the starting point for research of bipedal walking systems. This discovery of ZMP has initiated humanoid robotics development which is still ongoing. The consequence of such rapid development was that in some papers imprecise or erroneous interpretations of ZMP notion appeared. However, to point out misinterpretations and offer a correct understanding of ZMP, some papers[1–3] were published. However, insufficiently precise interpretations still sometimes appear, and we will try to illustrate the ZMP notion with extremely simple example, and add some explanations aimed at closing this issue.

In the shortest way — ZMP is an indicator of how well forces acting on biped are balanced to prevent the occurrence of the system falling down.

Let us imagine a weightless structure standing on a flat ground surface, as sketched in Fig. 1. Mass m has been placed in different positions on the structure. Because there is no other force acting on the structure — ground reaction force (GRF) will balance weight $\mathbf{G}$ with the force of the same intensity, acting along the same line, but in opposite direction. In cases (a) and (b), GRF is pointed vertically upward directly

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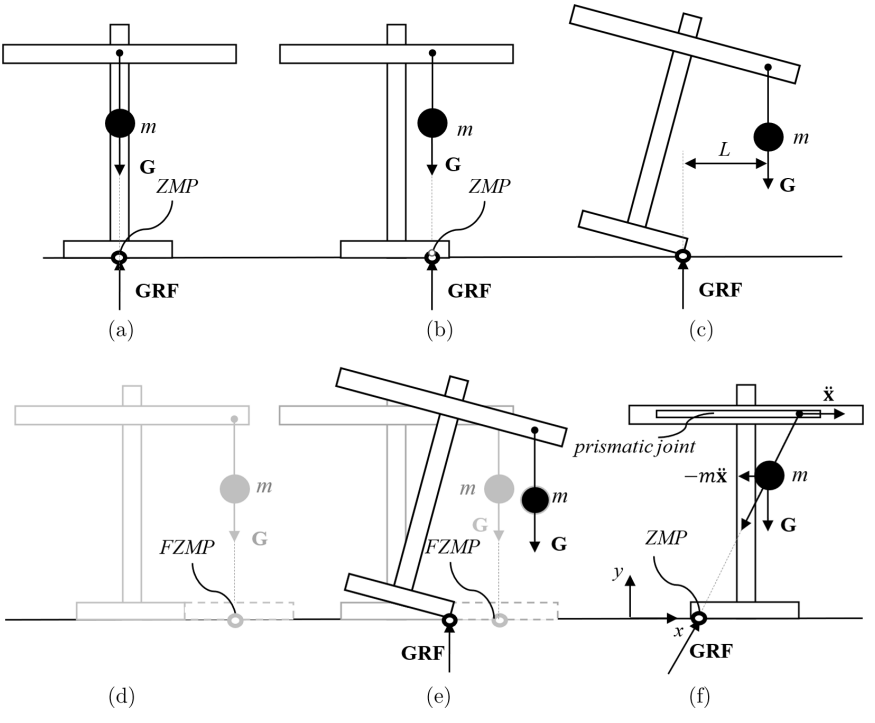


Fig. 1. Definition of ZMP.

below weight **G** (as visualized by a thin line) and in both cases was placed inside a support polygon. In both cases point at which GRF is acting is ZMP.

If mass m was placed such that weight vector **G** penetrates the ground surface out of support polygon [Fig. 1(c)] situation is different. Because weight was moved to the far right it can be expected that GRF will move in this direction according to position of **G**. However, GRF cannot move out of the support polygon (reaction force has to act on the weightless structure), and the situation as sketched in the case shown in Fig. 1(d) will happen. The system will incline and start rotating about the edge of the foot which remains in contact with the ground. The rotation torque about this edge will be $\mathbf{M} = \mathbf{L} \times \mathbf{G}$, and the GRF acting point is no longer ZMP because at this point perturbation moment is acting (moment is not zero), and causing the system to overturn. But in all cases, the GRF acting point is the Center of Pressure (CoP). This clearly shows that, in general, ZMP and CoP are not the same. They coincide only within the support polygon.

From Fig. 1 is clear that ZMP always should be within the support polygon. Now we will try to additionally clarify from where the claim that ZMP "can come out" of support polygon comes from. In case the ZMP position is not sensed by foot sensors but calculated, the calculation of ZMP position is done under the assumption that the system is not overturning, i.e., that the lowest part of the system (foot) is in full

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contact with ground. This can be done if the support polygon is large enough to encompass the obtained GRF acting point [foot size for this case is in Fig. 1(d) shown by dashed line]. However, this means that such calculated GRF acting point position is out of real support polygon and that the system overturn already started. This point in Ref. 3 has been named fictitious ZMP (FZMP), as shown in Fig. 1(d). In reality, GRF is acting on a foot edge about which the system rotates [now GRF acting point is not the ZMP, as shown in Fig. 1(e) and, for this case, ZMP does not exist].

During bipedal walking, body weight **G** might not be the only force GRF has to counteract. Let's suppose that mass m can have acceleration as sketched in Fig. 1(f). Force composed of **G** and inertial force has to be balanced by GRF. If the total force acting on the weight penetrates the ground surface within the support polygon, the penetration point will represent ZMP [Fig. 1(f)]. However, if the penetrating point is out of support polygon the system will overturn and eventually fall. Therefore, it is clear that all forces should be well balanced and canceled to ensure a "dynamically balanced walk".

We will not further elaborate on this topic and we point your attention to Refs. 1–3 for additional details.

In the rest of this paper, we will ensure more detailed insight into the two most important topics related to motion: motion in an unstructured environment and motion in the presence of additional contact above the ground surface. At the end, we will discuss the issue of walking learning.

3. Generalizing ZMP to Cases with Multiple Spatially Distributed Contacts

During walking, the humanoid can contact objects from the environment with the hand or some other part of the body (e.g., lean against the wall, hold on to the handrail, etc.) which imposes additional restrictions on possible movements in the presence of these contacts. Because the original ZMP notion was limited only to walking on flat surface, several authors have tried to generalize the concept of ZMP in the case of more spatially distributed contacts during movement.

Roles of spatial contacts may be of two types: to help to preserve dynamic balance and to enable motion to be continued while the contacts are preserved. Three tasks relating to contacts are of basic importance: contact maintenance, contact relocation while preserving dynamic balance and how to select suitable positions for new contacts.

If a humanoid is walking over a flat surface, dynamic balance will be endangered if the complete system rotates about the support polygon edge. A similar approach was adopted in Ref. 4 to study the balance of a humanoid robot having hand interaction with the environment. The authors formed a three-dimensional (3D) convex polyhedron of all supporting points (hands touching environment + support polygon on the ground) and investigated if the system can rotate about the polyhedron edge and

overturn under the influence of all acting forces. They introduced the term generalized ZMP (GZMP) as a point on the ground surface where the moment generated by reaction force and moment is zero. The reaction force and moment are generated by the inertial force and the gravity force. If GZMP is within the region obtained by convex polyhedron projection on the ground dynamic balance will be preserved.

In Ref. 5, the authors claim that a universal stability criterion of the foot contact of legged robots has been proposed when a robot walks on an arbitrary terrain. The robot's hands may, also, be in contact with the environment.[b] The proposed method checks if the sum of the gravity and the inertia wrench applied to the Center of Gravity (CoG) of the robot is inside the polyhedral convex cone of the contact wrench between the feet of a robot and its environment. However, just two examples (hands were not contacting the environment) were presented: biped walking on a horizontal plane and on stairs (in both cases sufficient friction was assumed). So, any advantage of taking into account spatial contact was not presented.

The ZMP notion assumes that all contacts with the environment are coplanar but even in the case of walking, contacts of feet and ground should not be on a single ground plane. Thus, one of the ideas to generalize ZMP notion was initiated by the idea presented in Ref. 6. "The generalization of the ZMP concept would be actually complete if we could define what is the pseudo-support-polygon, a certain projection of the three-dimensional (3-D) convex hull (built from the two real support areas) onto the virtual surface, inside which the pseudo-ZMP stays". This was the basis for the authors in Ref. 7 to propose a method for checking and enforcing multi-contact stability based on the ZMP. The key to this development is the generalization of ZMP support areas to take into account frictional constraints and multiple non-coplanar contacts. First, they proposed a method to derive the area that contains all ZMPs generated by valid contact forces (full support area). Then, the Linear Pendulum Mode (LPM) was applied to regulate the Center of Mass (CoM) altitude and the angular momentum to constant values. In this mode, CoM dynamics are related to the ZMP. As a consequence, constraints arising from the LPM shrink the support area (named pendular support area). Then, a whole-body controller was designed. A simulation of the HRP-4 humanoid locomotion across arbitrary multi-contact stances was performed.

An interesting approach was presented in Ref. 8 where spatial contacts were included in ZMP notion. The basic idea behind this approach is the following: if the humanoid is dynamically balanced, the external forces acting on the humanoid (including contacts) should produce the torque needed to balance its gravitational force and accelerate its CoM in the observed direction with the observed acceleration. To enable easy testing if this is fulfilled, they developed the Zero Point Line (ZML) notion and introduced a modified notion of support polygon.

[b]It was assumed that sufficient friction exists at each contact. This assumption implies that an arbitrary friction force can be generated at every contact point independently of the normal force at the point.

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They started with a simplified humanoid planar model — an inverted pendulum where a humanoid is approximated only by its CoM on which acts only the GRF and no torques, and rotations are present. From this follows — "in the ZMP, the sum of all external torques acting on the system and the torques generated by the external forces acting on the system equals the torque generated by the force acting in the COM. Thus, all torques in the ZMP cancel each other and their sum is zero". Then, authors come to ZML notion. Besides standing on the ground, humanoids can be supported by hands on different heights. At each contact — CoP has been formed and line connecting them was named CoP line. The authors concluded that if the system is dynamically balanced then "the sum of all existing torques in all COPs on any horizontal plane at any height is zero. In this case, all COPs are ZMPs, and the COP line becomes the ZML and passes through the support polygon of the investigated system". Then, they introduce a modified notion of support polygon. As the system can be spatially supported at different heights, the support polygon becomes a spatial region, extending between the outermost points where the support mechanisms are acting. However, the side of a system around which the support polygon extends depends on the orientation of the supporting forces (pulling or pushing). For different cases, they sketched different support polygons as lines in the sagittal plane. Finally, they concluded: if the ZML line passes through the humanoid and its support polygon — the humanoid is dynamically balanced. Authors claim that this can be applied to any system, even if its dimensions, the locations, and the acceleration of its segments are not known. Only the external forces, torques, and their locations should be known. To verify the approach two experiments with humans were performed.

In Ref. 9, authors presented a task where the robot have to move within the environment (completely known) by establishing and breaking multiple, non-coplanar contacts. The task is specified as a desired final stance that the robot must reach from its initial stance to which the initial configuration (set of joint angles) and contact wrenches are associated. Contact coordinate frames are attached to the robot at places that potentially may come in contact with the environment. When contacts are established, their positions and orientations are known. The authors presented a multi-contact planning and control framework which consists of two layers: planning and control layer. The planning layer works offline using two modules: the stance planner finds a sequence of stances, together with associated transitions and contact wrenches, and then the whole-body planner computes the sequence of motions to realize them. The control layer produces the torque commands allowing the humanoid to execute the planned motions while guaranteeing closed-loop balance. They tested the framework in the experiment with the COMAN+ humanoid robot. To robot was assigned the task to walk through a narrow and low passage (robot must walk as quadruped) and stand up in front of a wall which was just after the passage.

3.1. *Generalization for planar case*

In this section, we will try to get to a general dynamic balance constraint applicable to configurations with spatially distributed contacts and limited friction. We will

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start with a planar case and show how it generalizes to a spatial case. Much of this procedure has been described in Ref. 10 but we will briefly describe it here for the sake of completeness.

To reach the final form of dynamic balance constraint we have to start from analyzing simple point contact and it's properties. If the Coulomb friction is assumed, it is well known that frictional force lies within the friction cone as shown in Fig. 2. Since that is a quadratic constraint, we can approximate it with a set of linear constraints forming a friction pyramid. That constraint can be given in face form

$$\mathbf{S}_{\mu\, m\times 3}\mathbf{F}_{3\times 1} \succeq \mathbf{0}_{m\times 1}, \tag{1}$$

$$\mathbf{S}_{\mu k} = \left[\sin\frac{2k\pi}{m} \;\; -\cos\frac{2k\pi}{m} \;\; \mu\right]; \quad k = 0, \ldots, m-1, \tag{2}$$

where $\mathbf{S}_{\mu k}$ represents a single surface of the friction pyramid shown in Fig. 2. The alternative form is the span form given by

$$\mathbf{F} = \mathbf{u}_1\alpha^1 + \mathbf{u}_2\alpha^2 + \cdots + \mathbf{u}_m\alpha^m = \mathbf{U}\boldsymbol{\alpha}, \quad \alpha^i \geq 0, \tag{3}$$

where

$$\mathbf{u}_i = \begin{bmatrix} \mu\cos\dfrac{2\pi(i-1/2)}{m} \\ \mu\sin\dfrac{2\pi(i-1/2)}{m} \\ 1 \end{bmatrix}, \quad \mathbf{U} = [\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \ldots \;\; \mathbf{u}_m] = \begin{bmatrix} \mu\mathbf{C} \\ \mu\mathbf{S} \\ \mathbf{1} \end{bmatrix}, \tag{4}$$

and $\boldsymbol{\alpha} = [\alpha^1 \;\; \alpha^2 \;\; \ldots \;\; \alpha^m]^T$. Each $\mathbf{u}_i$ represents one side of the friction pyramid. In this representation, the total force is given as a linear combination of sides of the friction pyramid. Row vectors $\mathbf{C} = \left[\cos\frac{\pi}{m}, \cos\frac{3\pi}{m}, \ldots, \cos\frac{2m\pi-\pi}{m}\right]$, $\mathbf{S} = \left[\sin\frac{\pi}{m}, \sin\frac{3\pi}{m}, \ldots, \sin\frac{2m\pi-\pi}{m}\right]$ and $\mathbf{1}$ represent vector of cosine, sine and ones, respectively.

Let's consider a case when the robot is in contact with the ground with both of his feet and a wall with his hand as in Fig. 3(a). The feet are touching surfaces of different inclinations, and only that renders the ZMP notion inapplicable. As stated

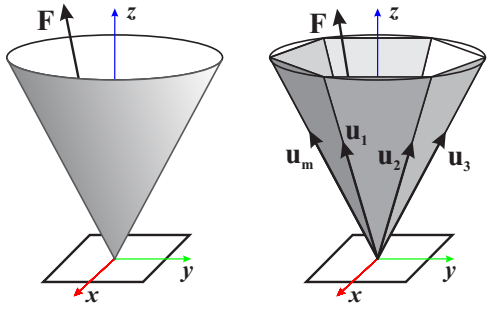
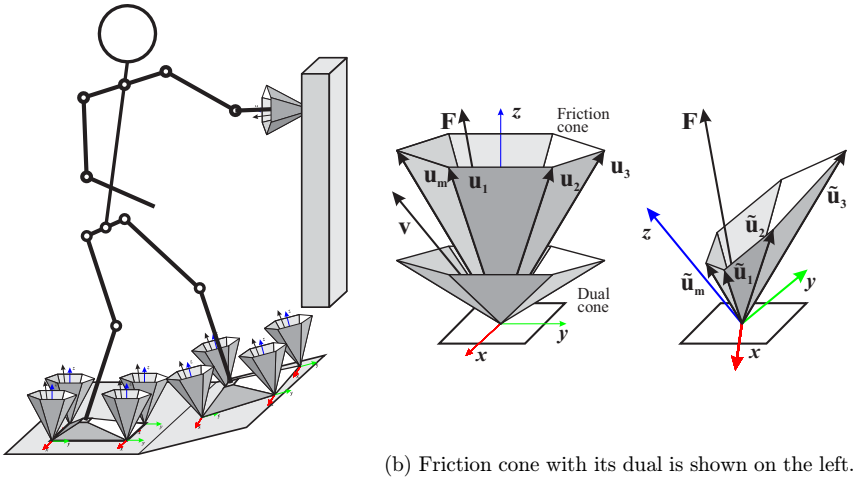


Fig. 2. The friction cone (left) and the approximating friction pyramid with generating vectors (right).



(a) A system with multiple contacts

(b) Friction cone with its dual is shown on the left. The modified friction cone is shown on the right

Fig. 3. Point contact and system with multiple spatially distributed contacts.

in Ref. 7, the surface contact can be modeled as a series of point contacts placed along the perimeter of the contact patch. If the contact patch is of a polygonal shape, the pyramids would be placed in the corners of the polygon.

The total wrench acting on any reference point $\mathbf{C}$

$$\begin{bmatrix} \mathbf{F}^{\mathbf{C}} \\ \mathbf{M}^{\mathbf{C}} \end{bmatrix} = \sum_{i=1}^{n} \begin{bmatrix} \mathbf{R}_i \mathbf{F}_i \\ [\mathbf{r}_i]_\times \mathbf{R}_i \mathbf{F}_i \end{bmatrix} = \begin{bmatrix} \mathbf{R}_1 \mathbf{U}_1 & \cdots & \mathbf{R}_n \mathbf{U}_n \\ [\mathbf{r}_1]_\times \mathbf{R}_1 \mathbf{U}_1 & \cdots & [\mathbf{r}_n]_\times \mathbf{R}_n \mathbf{U}_n \end{bmatrix} \begin{bmatrix} \boldsymbol{\alpha}_1 \\ \vdots \\ \boldsymbol{\alpha}_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Upsilon} \\ \boldsymbol{\Gamma} \end{bmatrix} \mathbf{A}, \tag{5}$$

where n is the number of separate contact points, $\mathbf{R}_i$ is the rotation matrix and $\mathbf{U}_i$ is the generating matrix for the ith contact. Vectors $\mathbf{r}_1$ to $\mathbf{r}_n$ represent positions of contact points relative to CoM of the system. $[\bullet]_\times$ is a skew-symmetric cross product operator matrix. Matrix $\boldsymbol{\Upsilon}$ represents the matrix where all generating matrices are stacked horizontally, $\boldsymbol{\Gamma}$ represents the matrix where all cross products of position vectors and generating matrices are stacked horizontally and $\mathbf{A}$ represents the vector of $\boldsymbol{\alpha}_i$s for all n contacts. In the rest of the paper, ith columns of matrices $\boldsymbol{\Upsilon}$ and $\boldsymbol{\Gamma}$ will be denoted as υ_i and $\boldsymbol{\gamma}_i$, respectively. The choice of reference point $\mathbf{C}$ is arbitrary. When analyzing whole-body dynamics, reference point will coincide with CoM. When analyzing surface contact stability point $\mathbf{C}$ will be some point on contact surface.

Let's first consider a case where all contact surfaces are parallel, meaning z axes of all friction pyramids coincide. Without the loss of generality, let's then assume all rotation matrices are identity and all friction cone generating vectors $\mathbf{U}_i$ are the same. It's a very relevant simplification because it describes a robot walking on a single inclined or flat surface or walking up and down the stairs. In such a case,

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Eq. (5) reduces to

$$\begin{bmatrix} \mathbf{F}^C \\ \mathbf{M}^C \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} \mu \mathbf{C} \boldsymbol{\alpha}_i \\ \sum_{i=1}^{n} \mu \mathbf{S} \boldsymbol{\alpha}_i \\ \mathbf{1} \sum_{i=1}^{n} \boldsymbol{\alpha}_i \\ \sum_{i=1}^{n} (y_i \mathbf{1} - z_i \mu \mathbf{S}) \boldsymbol{\alpha}_i \\ \sum_{i=1}^{n} (z_i \mu \mathbf{C} - x_i \mathbf{1}) \boldsymbol{\alpha}_i \\ \sum_{i=1}^{n} (x_i \mu \mathbf{S} - y_i \mu \mathbf{C}) \boldsymbol{\alpha}_i \end{bmatrix}. \tag{6}$$

From this equation, it is obvious that vertical force is the sum of all α's $F_z^C = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i^j$. Since $\alpha_i^j \geq 0$, it's easy to conclude that $F_z^C \geq 0$. If we look at other components of the wrench normalized by vertical force F_z^C we get

$$\begin{bmatrix} F_x^C/F_z^C \\ F_y^C/F_z^C \\ M_x^C/F_z^C \\ M_y^C/F_z^C \\ M_z^C/F_z^C \end{bmatrix} = \underbrace{\begin{bmatrix} \mu\mathbf{C} & \cdots & \mu\mathbf{C} \\ \mu\mathbf{S} & \cdots & \mu\mathbf{S} \\ y_1\mathbf{1} - z_1\mu\mathbf{S} & \cdots & y_n\mathbf{1} - z_n\mu\mathbf{S} \\ z_1\mu\mathbf{C} - x_1\mathbf{1} & \cdots & z_n\mu\mathbf{C} - x_n\mathbf{1} \\ x_1\mu\mathbf{S} - y_1\mu\mathbf{C} & \cdots & x_n\mu\mathbf{S} - y_n\mu\mathbf{C} \end{bmatrix}}_{5\times(nm)} \underbrace{\begin{bmatrix} \frac{\boldsymbol{\alpha}_1}{F_z^C} \\ \frac{\boldsymbol{\alpha}_2}{F_z^C} \\ \vdots \\ \frac{\boldsymbol{\alpha}_n}{F_z^C} \end{bmatrix}}_{(nm)\times 1} = \mathbf{P}\hat{\mathbf{A}}. \tag{7}$$

If we look at the elements of vector $\hat{\mathbf{A}}$, we can easily conclude that all of them are positive and sum up to 1. That means, that $[F_x^C/F_z^C, F_y^C/F_z^C, M_x^C/F_z^C, M_y^C/F_z^C, M_z^C/F_z^C]^T$ is a convex combination of columns of matrix $\mathbf{P}$, meaning that *it has to lie within a convex hull of points in 5D space, where each point is defined by a column of matrix* $\mathbf{P}$. Each hyperplane of the convex hull introduces an equation in the form

$$v_1 \frac{F_x}{F_z} + v_2 \frac{F_y}{F_z} + v_3 \frac{M_x}{F_z} + v_4 \frac{M_y}{F_z} + v_5 \frac{M_z}{F_z} \geq w. \tag{8}$$

After stacking all the equations into a single matrix form we can get

$$\mathbf{Z} \begin{bmatrix} \mathbf{F}_P \\ \mathbf{M}_P \end{bmatrix} \succeq 0, \tag{9}$$

where $\mathbf{Z}$ represents the dynamic balance matrix (DBM).

3.2. *Applicability of DBM*

As previously stated ZMP notion is applicable only in cases where all contacts are in the same plane, and there is enough friction. One of the previous works[11] considers a

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case with limited friction, but it only considers a rectangular foot with a four-sided pyramid as an approximation of the friction cone.

The DBM doesn't have these limitations, it takes into account limited friction and the number of faces in an approximation of friction cone is unlimited. To illustrate the method we have computed DBM for a rectangular foot using 4-, 8- and 12-sided friction pyramids (first, second and third column). Since DBM gives a constraint in 5D space it's hard to visualize the results. To do so, we have visualized three different cases, contact force is only vertical F_z, has x component $F_x = 0.5F_z$, and contact force has both x and y component $F_x = F_y = 0.5F_z$. It is visualized along with max vertical torque M_z for a given point, which is color-coded, where red color represents the highest max vertical torque and blue lowest. We have visualized the points on the ground surface where tangential torque is zero. The results are visualized in Fig. 4. That point will be named extended ZMP or eZMP. It closely relates to ZMP with one subtle difference. The ZMP is defined only for planar case, and in then eZMP coincide. When we have a spatial case as in, e.g., Fig. 5(c) the support surface is not well defined and ZMP doesn't exist. The area where eZMP can lie while still guaranteeing dynamical balance is shown. It is important to note here that the area where eZMP guarantees dynamical balance is the same as the ZMP support area since ZMP is well defined. Such an area will be called a eZMP support area.

It is easy to note that when the approximation is too coarse, some areas with high max friction available might disappear. In reality, that means when using DBM constraint to test for the contact stability we will be getting false negatives. The test

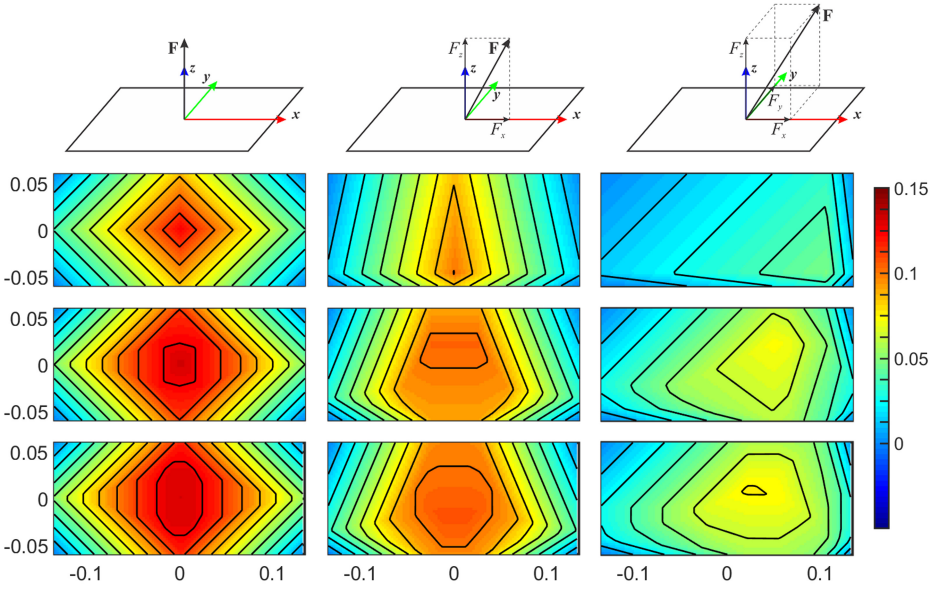


Fig. 4. Influence of forces and approximation quality on the effective shape of foot.

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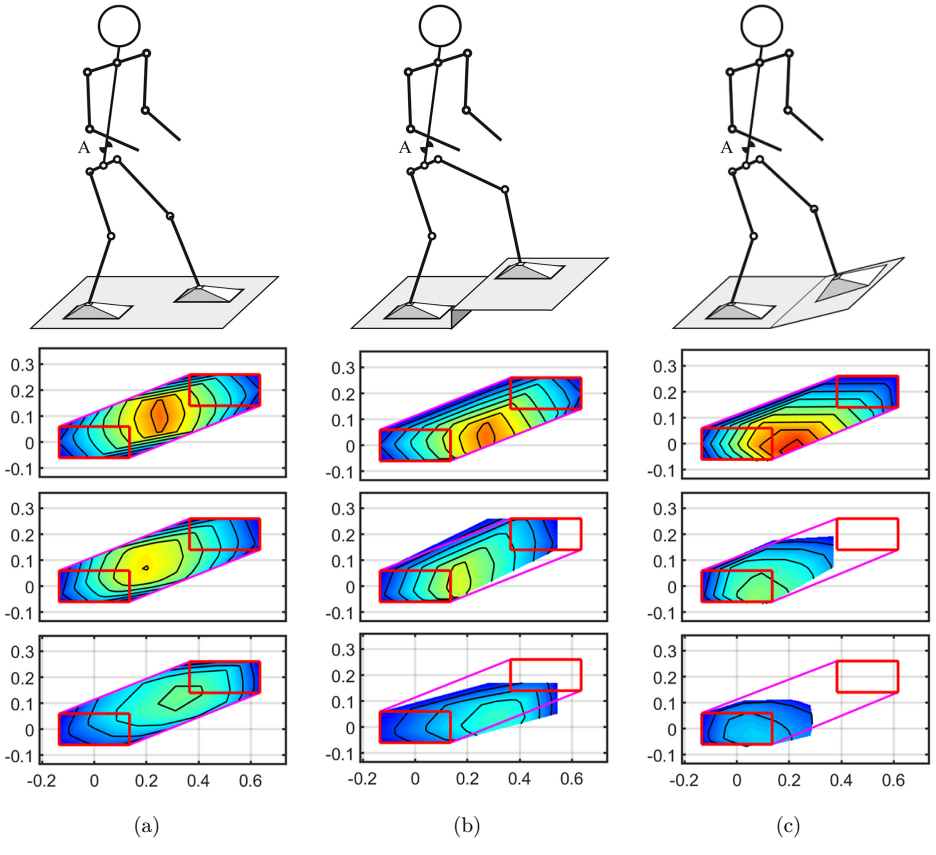


Fig. 5. Three typical walking configurations. Walking on flat ground, walking up the stairs and stepping on inclined surface. Eight-sided pyramids are used for approximation. Graphs show eZMP support area given contact configuration, horizontal forces and vertical torque. First row is case where only F_z is present, second row has $F_x = 0.5F_z$ and third row is case with $F_x = F_y = 0.5F_z$.

would indicate that contact is unstable although it is stable under a given contact wrench.

The second thing to note is that external forces are modulating the effective shape of the foot. The black lines in Fig. 4 represent the effective shape of the foot given F_x, F_y and M_z. If the force is in x direction the high M_z area will move downwards. If we have force in both x and y direction it will move bottom-left. In some extreme cases only a small fraction of the whole contact patch guarantees contact stability.

In Fig. 5, we have three typical stepping cases, walking on a flat ground, walking up the stairs and stepping on the inclined surface. With tools developed so far, we can compute wrench constraints for the first two cases, while for the third case, we must wait a bit more. Unlike in the previous case, now we have a double support phase and eZMP has to be within the convex hull of the contact patch in the classical ZMP notion [Fig. 5(a)]. Unlike commonly thought, when stepping up the stairs the feasible

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eZMP area is contact patch and is not simply a convex hull of projection of feet to the ground plane. In the presence of tangential forces F_x and F_y, the eZMP can lie outside of the projected contact patch but still guarantee dynamical balance. In the second and third cases, the ZMP is not defined, since we don't have a single contact surface, hence we use eZMP notion.

3.2.1. *Non-coplanar contact case*

As we have seen in the previous section, DBM can be applied to general cases where the only assumption is that all contact surfaces are parallel and that z axes of all friction pyramids are the same. In cases shown in Figs. 3(a) and 5(c) such an assumption doesn't hold. We can reduce the spatial case to the planar one if we can find a unit vector $\mathbf{v}$ such that

$$\mathbf{v}^T \mathbf{R}_i \mathbf{u}_i^j \geq 0, \quad i \in 1, \ldots, n, j \in 1, \ldots, m. \tag{10}$$

That condition is equal to $\mathbf{v}^T \boldsymbol{\Upsilon} \succeq 0$. Such a vector $\mathbf{v}$ would lie within the dual cones[c] of all friction cones. It is possible to compute rotation matrix $\mathbf{R}_v$ that aligns new z axis with a vector $\mathbf{v}$. An example of such a rotation is given in Fig. 3(b). After that, all friction pyramids can be expressed in the new coordinate frame where z equals v

$$\tilde{\mathbf{u}}_i^j = \frac{\mathbf{R}_v \mathbf{R}_i \mathbf{u}_i^j}{\mathbf{v}^T \mathbf{R}_i \mathbf{u}_i^j}. \tag{11}$$

The cone is normalized so z component of vector $\tilde{\mathbf{u}}_i^j$ is 1 as in (4). Since the denominator of the previous equation is positive as by (10), that original and modified friction pyramid generates the same space. The reference point for which we compute the constraint is arbitrary, but in this case, the notion of contact patch and contact surface is unclear. Hence, the reference point will be the CoM of the object. Hence, the constraint on the wrench acting on CoM gives the constraint on centroidal motion. If we compute the total wrench acting on CoM we get

$$\begin{bmatrix} \mathbf{F}^{\mathbf{v}} \\ \mathbf{M}^{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{U}}_1 & \cdots & \tilde{\mathbf{U}}_n \\ [\mathbf{r}_1^v]_\times \tilde{\mathbf{U}}_1 & \cdots & [\mathbf{r}_n^v]_\times \tilde{\mathbf{U}}_n \end{bmatrix}. \tag{12}$$

This is the same as form (5), meaning that we can use the same procedure to compute constraint on total wrench action on CoM as one described in Sec. 3.1. That way we can reduce the spatial case to a planar case, and use tools previously developed to compute a feasible CoM wrench. Feasible CoM wrench directly relates to the feasible centroidal motion of the system

$$\mathbf{W} \begin{bmatrix} \mathbf{F} \\ \mathbf{M} \end{bmatrix} = \mathbf{W} \begin{bmatrix} m(\mathbf{a} - \mathbf{g}) \\ \dot{\mathbf{L}} \end{bmatrix} \succeq 0, \tag{13}$$

[c]Dual cone of cone C in 3D space is defined as $C^* = \{\mathbf{y} \in \mathbb{R}^3 : \mathbf{y}^T \mathbf{x} \geq 0, \forall \mathbf{x} \in C\}$.

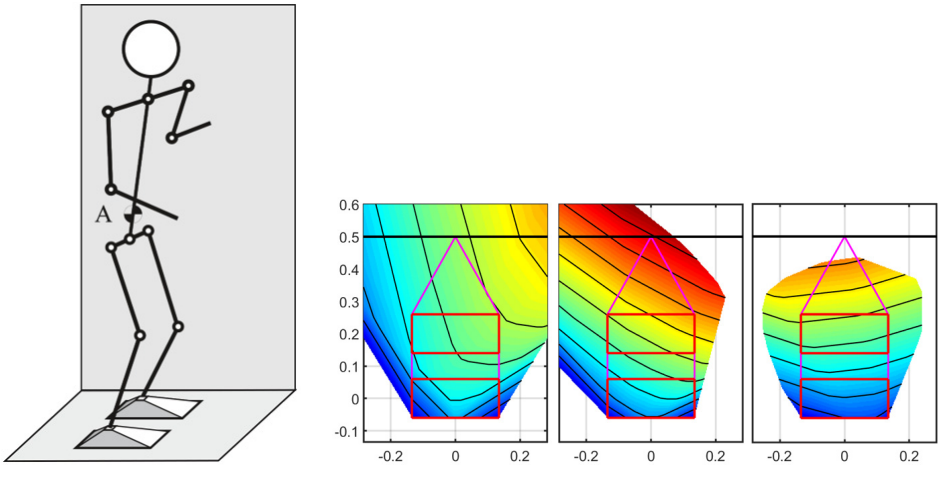
where **W** is the wrench constraint matrix, computed following the procedure described in Sec. 3.1. The difference between it and the DBM is very subtle, as DBM is defined only for planar cases and in some sense is analogous to ZMP, while wrench constraint matrix **W** is considered in the spatial case and reference point is CoM.

The results of this procedure are illustrated in Fig. 5(c) where one foot is in contact with the horizontal plane while the other foot is on the inclined surface. To illustrate the approach, we have once again reduced computed the position of eZMP that would guarantee the dynamical balance. It is worth noting, that if we want a robot to move forward, having $F_x > 0$, we can't rely on the front foot. That is clearly visible from Fig. 5(c) since the "eZMP support area" doesn't include the front footprint.

Another example is shown in Fig. 6, where a robot is touching the vertical wall with its hand. It is easy to note that the eZMP support area extends way above the convex hull of the projection of all contacts on the horizontal surface. This is, of course, an expected result, and adding a contact between the wall and hand extends the mobility of the robotic system. Of course, if the robot wants to move left, the [third graph in Fig. 6(b)], the hand contact doesn't help as much, since it can only generate forces in the opposite direction. That can be observed in the rightmost graph of Fig. 6(b), where the eZMP support area is smaller than one in the leftmost graph and doesn't contain a projection of left-hand contact.

3.2.2. *Shifting WCM*

As the robot moves around so does the constraint on the CoM defined by (13) change, since the position of contacts relative to CoM changes. As a result, WCM



(a) (b)

Fig. 6. Robot standing on the ground with additional contact between left arm and wall. In first graph case where only F_z is present, in the second graph $F_x = 0.5F_z$ in third graph $F_x = F_y = 0.5F_z$.

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needs to be updated. Fortunately, we don't need to run the same process at each timestep. The force $\mathbf{F}_B$ and torque $\mathbf{M}_B$ acting at position $\mathbf{r}_B$, would create force $\mathbf{F}_A = \mathbf{F}_B$ and torque $\mathbf{M}_A = \mathbf{M}_B + (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{F}_B$. Applying that to (13) yields

$$\mathbf{W}\begin{bmatrix}\mathbf{F}_A\\ \mathbf{M}_A\end{bmatrix} = \mathbf{W}\begin{bmatrix}\mathbf{I} & \mathbf{0}\\ [\mathbf{r}_B - \mathbf{r}_A]_\times & \mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{F}_B\\ \mathbf{M}_B\end{bmatrix} = \mathbf{W}^\Delta\begin{bmatrix}\mathbf{F}_B\\ \mathbf{M}_B\end{bmatrix} \succeq 0. \tag{14}$$

This way we derived a shifting rule for WCM. If the WCM is computed for some reference point if the reference point shifts by $\mathbf{\Delta}$, the WCM for the new reference point is

$$\mathbf{W}^\Delta = \mathbf{W}\begin{bmatrix}\mathbf{I} & \mathbf{0}\\ [\mathbf{\Delta}]_\times & \mathbf{I}\end{bmatrix}. \tag{15}$$

As a result, we need to compute WCM only once per contact configuration. As CoM moves, we can compute the shifted WCM by (15). It gives the same constraint on wrench produced by contact forces (13) as regular WCM but it's much faster to compute.

3.3. *Inability to reduce to planar case*

As stated, the previous procedure of reducing the spatial case to the planar case depends on our ability to find such unit vector $\mathbf{v}$, for which it holds $\mathbf{v}^T\mathbf{\Upsilon} \succeq 0$ [Eq. (10)]. The non-existence of such a vector would mean that there is no intersection between dual cones of all friction cones. An example of such a configuration is given in Fig. 7, where the robot is trying to climb between two vertical walls. For that case it is obvious that whatever the friction coefficient, there can't be an intersection between dual cones, and vector $\mathbf{v}$, complying to Eq. (10) can't exist. On the other

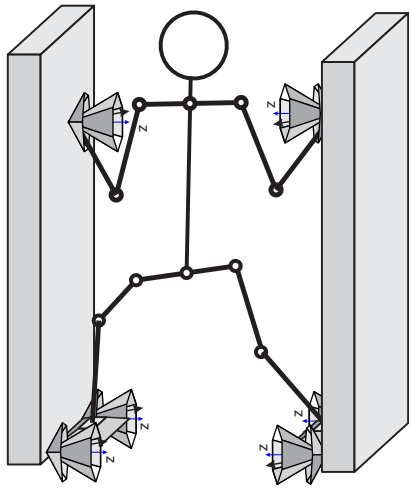


Fig. 7. Robot pushing against vertical walls. The nonzero intersection of duals of all six friction cones doesn't exist.

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hand, we are well aware that a person can stand and move in such a configuration, meaning there must exist a feasible CoM wrench.

To derive the constraint on motion we will use the well-known Farkas–Minkowski theorem,[12] stating that as follows.

Farkas–Minkowski. Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Then, exactly one of the following two assertions is true:

(1) There exists $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \geq 0$.
(2) There exists $\mathbf{y} \in \mathbb{R}^m$ such that $\mathbf{A}^T\mathbf{y} \geq 0$ and $\mathbf{b}^T\mathbf{y} \leq 0$.

If we apply that to Eq. (5), having in mind that in case we are discussing a $\mathbf{v} \in \mathbb{R}^3$ such that $\mathbf{v}^T\boldsymbol{\Upsilon} \succeq 0$ and $\mathbf{F}^T\mathbf{v} < 0$ doesn't exist, we can conclude that for arbitrary $\mathbf{F}$, we can find $\mathbf{A}$, such that $\boldsymbol{\Upsilon}\mathbf{A} = \mathbf{F}$. Thus, in this contact configuration, it is possible to generate arbitrary force at (any) reference point, including CoM. Hence, in such a case we can achieve arbitrary acceleration of CoM.

3.4. *Simulation results*

In this section, we will briefly describe simulation results obtained with the help of DBM and WCM. Following the previous discussion, we will illustrate two distinctive cases:

- The contact configuration cannot be reduced to a planar case.
- The contact configuration can be reduced to a planar case.

In both cases, we will use the generalized task prioritization framework.[13] Such a framework enables users to compute control torque based on a stack of tasks where task can be either of equality type or inequality type. Lower-priority tasks are performed so that execution of higher-priority tasks is not affected. The list of tasks is as follows:

(1) Joint torques have to be within predefined bounds.
(2) Joint angles have to be within predefined bounds.
(3) Contacts have to be stable.
(4) CoM has to follow a predefined trajectory.
(5) The limb that is repositioned has to follow a Cartesian trajectory.
(6) The robot's posture has to be as close to the initial posture as possible.

The list of tasks for both cases is practically the same, with the subtle difference, that when contact configuration can be reduced to planar case we have to pay special attention to CoM motion. In that case CoM motion is bounded and we have to take that into account when defining CoM trajectory. The contact patterns and CoM trajectories were generated manually, since that is out of scope of this paper. In all simulated cases, friction coefficient was assumed to be 0.8.

The first simulated case covers the situation where a robot is climbing between vertical walls as shown in Fig. 8. The robot alternates between being supported by

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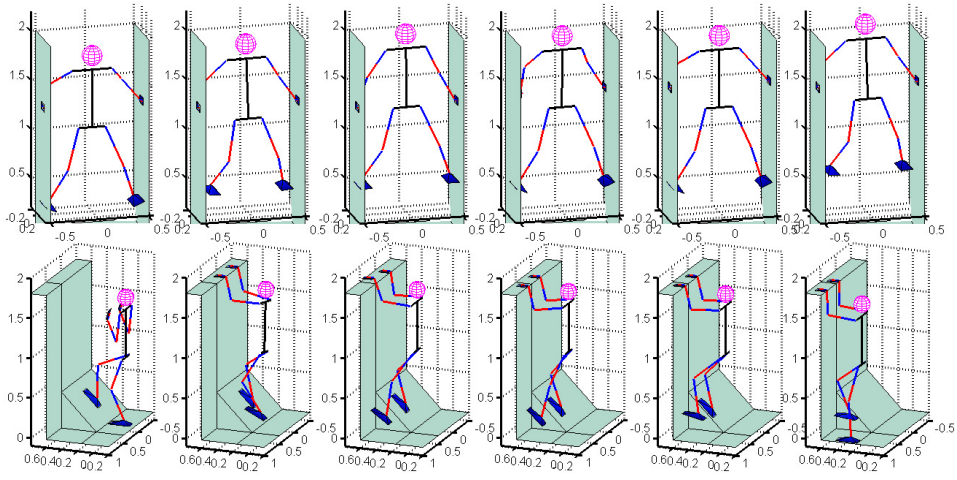


Fig. 8. Top row: A robot climbing between the vertical walls. Bottom row: Robot moves sideways while holding on the top of the wall.

feet while repositioning hands and being supported by hands while repositioning feet. As stated previously there is no constraint on CoM acceleration, and a robot is able to move upwards although there is no horizontal support. Although the contact configuration allows for such a motion, a high amount of joint torques is required. In our scenario, we assume that joint motors can provide sufficient torque, but if that's not the case robot won't be able to perform the motion, although it complies with WCM constraint.

In the second simulated case, the robot moves sideways on a wall while changing the contact configuration multiple times, as the motion progresses. It is illustrated in Fig. 8. It starts with one foot on the ground and one on a 45° slope. Since it's impossible to just lift the left leg, while maintaining zero CoM acceleration, which can be checked using WCM, the robot puts both hands on the top surface. After that, he puts his left leg in line contact with the vertical wall and then steps down. The amount of contacts, their positions and orientation change during the motion. The WCM computation was implemented in Python together with Numpy and CVXOPT packages, and it took on average 0.8 ms, 1.6 ms and 2.7 ms when 4, 8 and 12 friction cones are present. If the number of sides in the friction pyramids increases, the computation times would go up to 1.4 ms, 3.4 ms and 5.3 ms for six-sided and 2.3 ms, 4.9 ms and 8.0 ms for eight-sided pyramids. As stated above WCM needs to be computed only once per contact configuration.

4. Walking in Unstructured Environment

Humanoid biped robots designed to operate within human living and working environments cannot be pre-programmed due to the unpredictable nature of these

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settings. This unpredictability arises from the fact that the surroundings are not of a predetermined structure and are always changing. Consequently, it is impossible to have prior knowledge of the precise layout and positions of objects in the robot's immediate vicinity. As a result, it becomes imperative to make real-time decisions regarding the robot's motion parameters, such as walking direction and speed, as well as the generation and execution of the walking path.

Furthermore, the dynamic nature of such environments adds complexity. The ground surface may not always be perfectly flat and uniform, with features like stairs, varying ground levels, etc. Traditional wheels may not suffice for ensuring the required mobility in these situations. Instead, bipedal locomotion or movement on two legs, appears to be the most suitable approach. However, achieving bipedal locomotion is a challenging engineering task. Two simultaneous requirements must be met: the legs must be driven in a manner that facilitates the system's movement along a chosen path, and the entire body's motion must be coordinated in a synergistic manner to maintain dynamic balance consistently.

In modern humanoid robots, several approaches are used for the synthesis and generation of biped walk. The first such method, pioneered by Vukobratović and Juricic,[14] for synthesizing a dynamically balanced walking pattern was the semi-inverse method. In the semi-inverse method, the motion of legs is predefined, while the motion of the remaining parts of the system is determined in a way to ensure repeatability conditions and ensure dynamic balance during walking. This balance is achieved by keeping the ZMP at a specified position within the support area. Many subsequent methods introduced over time are essentially variations and enhancements of this fundamental approach.

Algorithms for walk generation that is based on simplified models, i.e., the linear inverted pendulum model (LIPM) of a biped robot,[15] use a predefined footstep sequence provided by the user to generate a feasible CoM trajectory. In an attempt to overcome issues associated with the strict assumptions of the LIPM, researchers have introduced more complex pendulum models. The largest constraint on LIPM walking is the assumption of constant CoM height, leading to the development of a nonlinear inverted pendulum with a variable mass height.[16] To account for the inertia of a swinging leg, in Ref. 17 is proposed the addition of a mass at the swing foot, which they termed the gravity-compensated LIPM. One of the most commonly used models in the literature to address nontrivial angular momenta from the limbs of large robots moving dynamically is to add a flywheel to the hip, which can be used to represent the inertia of the robot body as it moves.[18] A flywheel model of the robot has gained recognition as a convenient representation of the robotic system. Various pendulum models have been widely used in the analysis of push recovery and balance.

One of the widely utilized approaches for walk generation specifies the desired positions of the robot's feet on the ground[19,20] along the intended path, while ensuring the robot's dynamic balance. When footprint positions are defined on the ground surface, and the timing for the robot to reach these positions is established, it not only determines the direction of movement but also the speed. This data forms

the foundation for determining the foot trajectories, and by using inverse kinematics, it becomes possible to calculate the necessary joint trajectories to achieve the desired foot movements. Furthermore, based on the predetermined foot positions and the designated timing for reaching them, it is essential to establish a reference ZMP trajectory. To maintain the dynamic balance of the system, the motion of the remaining components of the system is computed to follow this reference ZMP trajectory.

As a result of the rapid developments within the trajectory optimization community, researchers began to move towards utilizing nonlinear dynamic gait optimizations rather than relying on the constraints imposed by linear modeling assumptions. Computation power finally increased sufficiently throughout the mid 2000's to begin handling 3D dynamic walking behaviors.[21] While nonlinear optimization methods do take into account the complete body dynamics of the robot, there is often an advantage to integrate gait generation and controller synthesis into a unified framework known as closed-loop optimization.[22] This approach offers several benefits, including the ability to generate walking behaviors that can be proven to be stable while simultaneously adhering to various system constraints, ranging from admissible configurations to torque limitations. The optimization process spans a single-step cycle, from heel strike to heel strike, and it is crucial to ensure that the motions align with the physical system's limitations, such as the friction cone, actuator boundaries and joint constraints.

Another approach, which will be presented in more detail in this section, introduces the generation of the walk by combining simple movements. When observing humans, walking itself seems to be a straightforward and easily achievable task. Neurological studies[23,24] proved the existence of specific leg movements that emerge as recurring patterns. Nonetheless, these patterns don't have to be rigidly uniform. The actual shape of these movements, which make up the pattern, often undergoes alterations and adjustments to adapt to immediate circumstances. If a robot possesses a humanoid structure that closely resembles that of a human, it becomes feasible to apply the same or similar concept to realize biped walking of humanoid robots. This inspired the development of a framework for generating walking patterns using basic simple movements, i.e., motion primitives,[25] that can be assembled into a parametrized walk, capable of swiftly and flexibly adapting to changes in the robot's environment.

4.1. *Synthesis and realization of walk composed of motion primitives*

The assessment of humanoid robot walking performance is often benchmarked against human walking capabilities. Nevertheless, humans acquire their walking skills from infancy, undergoing continuous refinement and adaptation to changes in their body's kinematics and dynamics. Over time, walking becomes a highly refined and automated process, with the underlying sensory-motor control system operating seamlessly in the background. Most individuals can effortlessly execute various types

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of walking, counteract external disturbances, adapt to different ground structures, and concurrently engage in additional tasks such as object manipulation, and more. When combined with sensory perception, primarily through vision, humans excel at navigating unfamiliar surroundings, reaching their objectives while adeptly navigating a variety of obstacles, including stairs, doors and other common elements from their environment.

When walking, humans don't pre-plan and place footprints on the ground before they commence walking. If we expect robots to function effectively in typical human environments, it's imperative that their walking abilities closely resemble those of humans. Adaptability to the current environmental conditions, allowing for real-time adjustments to the walking path and gait parameters, is essential. Moreover, given the ubiquitous presence of disturbances, the control system must possess robustness, speed and responsiveness to effectively compensate for these disruptions. Furthermore, path generation, encompassing approaches such as approaching, avoiding and circumventing obstacles, should draw inspiration from human methods too. Hence, the motion synthesis system should enable online modifications of planned paths and walking parameters, supported by a robust motion control system capable of compensating disturbances while maintaining dynamic balance consistently.

Walking synthesized with motion primitives represents an approach that allows for the execution of complex motions like walking without the need to pre-program joint trajectories for an entire step in advance. Instead, motion is defined by a set of parameters that specify the desired position of the last segment of the kinematic chain relative to the robot's base coordinate frame. This results in the coordinated movement of joints responsible for driving the last segment (such as a foot, head or hand) toward a parametric goal position. These parameters can be adjusted at any moment, allowing the synthesized motion to adapt online. The motion of limbs exhibits smoothness, including transitions between consecutive primitives. This smoothness is achieved by gradually altering the velocities of last segment toward their new goal positions. An example of a motion primitive within a walking cycle is leg stretching, which involves simultaneous motion of leg joints. This is followed by another motion primitive: leg bending, with a seamless transition of foot velocity toward the new target position. An essential aspect of realizing walking is the preservation of dynamic balance. While each motion primitive ensures appropriate trajectory shapes and smooth transitions from the preceding primitive, it does not inherently guarantee dynamic balance preservation. To meet these requirements, from path planning to execute motions at the joint level, we've proposed a cascade controller (Fig. 9).

The controller has four key blocks:

- The first block handles path planning, calculating the trajectory towards the goal position or adapting it in the case of changes in the environment.
- The second block determines desired joint angular velocities based on a combination of tied-up motion primitives.

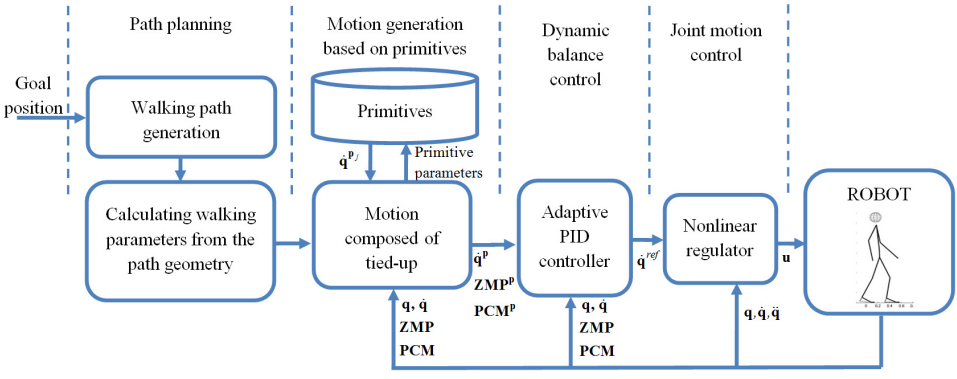


Fig. 9. Cascade walking controller composed of the modules for path planning, motion generation with motion primitives, dynamic balance preservation and joint motion control.

- Since motion primitives do not inherently account for dynamic balance, the third block introduces corrections to desired joint velocities, factoring in the current position of the ZMP and CoM projection to ensure dynamic balance preservation.
- The fourth block computes joint control values. Given the highly nonlinear and coupled nature of humanoid robots, this block encompasses feedback linearization, sliding mode control and disturbance estimation.

4.1.1. *Decomposition of walk*

This method of synthesizing walking allows for online adjustments, where overall motion parameters, including walk speed (W_{Speed}), leg height during the swing phase (W_{Height}), step length (W_{Length}) and walking direction (W_{Dir}), can be modified. How the relationship between these parameters and the parameters of the motion primitives is established, will be explained in detail in the following section.

Walk on a flat surface can be considered as composed of simple movements, similar to the previously described leg stretching and leg bending (Fig. 10). In total, there are five distinct primitives identified for leg movements, along with one for the trunk and one for the arms. The motion primitives for legs encompass (i) leg bending, (ii) leg stretching, (iii) inclining the robot forward during a single support phase, (iv) ensuring foot surface contact after the heel strike and (v) transferring body weight onto the subsequent supporting leg during a double support phase. Two additional primitives are (vi) maintaining an upright trunk posture and (vii) governing arm swing, a motion that counterbalances leg motion. Arm swinging serves to reduce the body's angular momentum and contribute to the preservation of dynamic balance.

Each motion primitive is responsible for defining the motion of one kinematic chain at a time, and two motion primitives for the same kinematic chain cannot be executed at the same time. Each motion primitive has its parameters, for example, leg bending has parameters — speed and height of the foot at the end of the primitive motion. These parameters are manually determined during the synthesis

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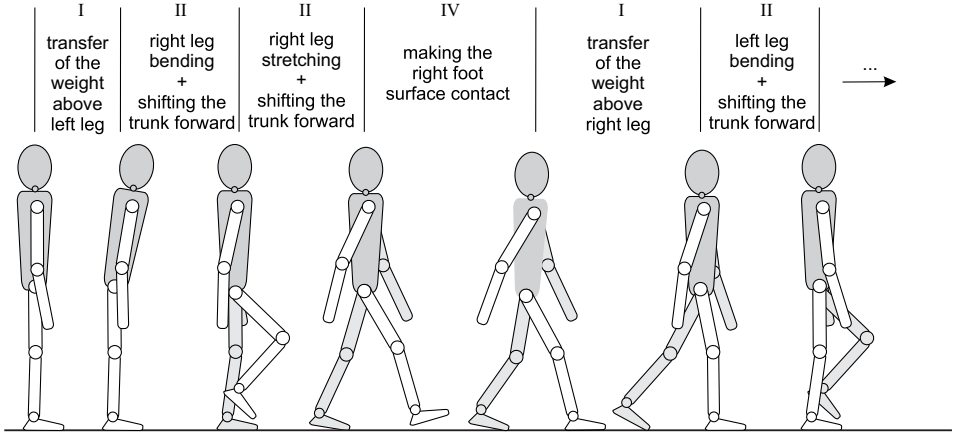


Fig. 10. Decomposition of walk into the motion primitives through four phases of walk.[25]

phase and can be fine-tuned and learned in different ways.[26] Moreover, every parameter within each motion primitive is contingent on walk parameters. The walking speed not only affects the pace of each motion primitive but for example it also influences the leg stretching parameter. In other words, a faster walking speed results in faster and longer steps.

4.1.2. *Preservation of dynamic balance*

A humanoid robot is under actuated, multivariable, coupled and highly nonlinear system. The gait a humanoid robot performed has to meet functional requirements, and it has to ensure dynamic balance simultaneously. The balance control approach we propose is centered around a zone surrounding the desired ZMP position, which changes at each time instant. Desired ZMP depends on current primitive and walking parameters. A zone around the desired primitive is introduced, and if the ZMP remains within this zone, the system is considered dynamically balanced. This indicates the ZMP is sufficiently close to the desired position, and the control system manages the robot's posture effectively. However, when the ZMP falls outside of this zone, the control focus shifts to the task of preserving dynamic balance. This controller switches between these two modes continuously and gradually. This control strategy is driven by two primary concepts. First, it aims to prevent situations where the system's dynamic balance is seriously compromised by applying corrections while deviations are still small. Second, it ensures that any corrections or switches between different control strategies are carried out smoothly to keep the ZMP within the support area, allowing for sustained and dynamically balanced biped walking.

The dynamic balance controller modifies joint velocities determined from motion primitives. Modifications are determined on the basis of the desired and current ZMP as well as on the current joint angles and angular velocities. One solution for this controller is an adaptive PID regulator whose detailed structure is given in Ref. 25.

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Modified joint velocities are to be executed in joints. For this step, it is necessary to calculate control variables (joint torques or motor voltages). Due to the properties of humanoid robots, one possible solution for a joint-level controller can be a combination of feedback linearization, sliding mode controller and disturbance estimator.

The dynamic balance controller is responsible for adjusting joint velocities derived from motion primitives. These modifications are based on a combination of different signals, including the desired and current ZMP along with the current joint angles and angular velocities. One potential solution for this controller is the implementation of an adaptive PID regulator, the detailed structure of which is outlined in Ref. 25. The adjusted joint velocities need to be executed in the robot's joints, requiring the calculation of control variables (joint torques or motor voltages). Given the characteristics of humanoid robots, a viable approach for the joint-level controller could involve a combination of feedback linearization, sliding mode control and a disturbance estimator.

4.2. *Walking in the unstructured and dynamic environments*

Considering the dynamic and unstructured nature of the human environment, it is crucial to enable real-time modification of the robot's walking path for efficient navigation. Walking patterns generated from primitives offer adaptability due to their inherent relationship with walk parameters. This relationship facilitates automatic adjustments of primitive parameters by imposing different walking conditions. The parameters governing basic walking were determined empirically. A more precise definition — learned values or fine-tuned values of primitive parameters result in more anthropomorphic motion.

Bearing in mind that the characteristics of the path define the way of walking, a strong relationship between the parameters of the path and the walking parameters is established. The four key walking parameters (W_{Speed}, W_{Dir}, W_{Height} and W_{Length}) can be derived from the geometric attributes of the path. Consequently, an approach is introduced in Ref. 27 for generating the path and calculating walking parameters based on its geometry.

An important inquiry pertains to the independence of the four walking parameters or whether there exist relationships among them. Additionally, the shape of the human walking should be addressed. To achieve anthropomorphic robot walking, insights from physiology and biomechanics were incorporated. In Ref. 28, various strategies were explored, emphasizing the formation of a walking path with a predefined goal position. It was assumed that the path selection adheres to optimization principles, and the results indicated that the cost function to be minimized is a time derivative of path curvature. This finding led us to the adoption of clothoid arcs as a suitable approximation for the path shape in our path-planning algorithm.

Once the desired goal position relative to the robot's current position has been defined, the utilization of a clothoid curve presents itself as a suitable method for interconnecting these two sets of coordinates. At any given moment, it becomes

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possible to establish a new goal point, thereby generating a new clothoid arc that determines the robot's path. This dynamic capability allows for real-time adjustments to the robot's path in response to unforeseen or dynamic obstacles while ensuring a seamless transition from the previous to the newly generated trajectory.

In our work presented in Ref. 27, we have implemented a forward goal-oriented locomotion approach, without the inclusion of backward or lateral steps. For this level of control, the locomotion of the robot is approximated as a non-holonomic unicycle model, which introduces a relationship between the walking direction and the tangent to the generated path. This interdependence serves as a basis for the calculation of one of the key walking parameters, i.e., the walking direction W_{Dir}. Studies of gait physiology and biomechanics have provided confirmation of the existence of a relationship between stride length and cadence, contributing to the automatic gait control mechanism. Stride length is precisely defined as the distance between successive heel strikes of the same foot within the walking plane. This relationship exhibits linearity when individuals walk at their self-selected speed. Consequently, an increase in walking speed entails longer strides, but this relationship reaches a breakpoint when cadences exceed 150 steps per minute. For faster walks, stride length reduces with further increases in cadence.

Given that walking speed is the product of cadence and stride length, we can establish a clear connection between two essential walking parameters, i.e., W_{Speed} and W_{Length}. Hence, the definition of one of these parameters suffices, as the other can be derived accordingly. Further analysis of walk parameters introduces a relationship between path curvature and walking speed, known as the $1/3$ power law. While this relationship has traditionally been considered valid some studies proved the coupling between velocity and curvature is not universally applicable. Given that our algorithm primarily employs clothoid path shapes, and considering the ongoing debate in the literature, we have adopted the $1/3$ power law as the governing relationship between path curvature and walking speed, from which we can calculate two aforementioned parameters W_{Speed} and W_{Length}.

The fourth key walking parameter, W_{Height}, remains independent of other parameters and is unaffected by the geometric properties of the path. Instead, it depends solely on the estimated height of any encountered obstacles. The presence of such obstacles or changes in terrain conditions (such as approaching stairs) exerts an influence on the remaining walking parameters, that require a robot to be prepared for the transition between different gait types. An example of a robot walking around obstacles while following the path generated with clothoid curves is shown in Fig. 11.

The generation of clothoid curves takes as input two values: the current position of the robot and the designated goal position. However, when an obstacle moves and potentially obstructs the direct path between these points, we need an algorithm for selecting intermediary waypoints to circumvent these obstacles. One such algorithm is as follows. It computes the shortest clothoid path between the robot's current location and the target goal position. The computation of these waypoints takes into account the robot's present position and the visible portions of the obstructing

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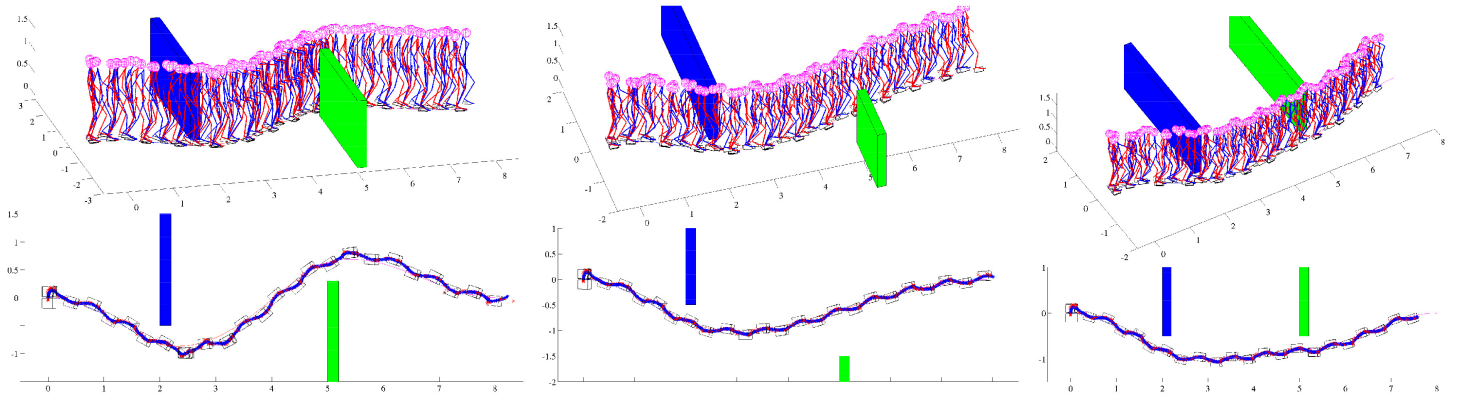


Fig. 11. Walking in the environment with the different obstacle configurations: Stick diagram of robot motion (top) and robot footprints with ZMP (down).

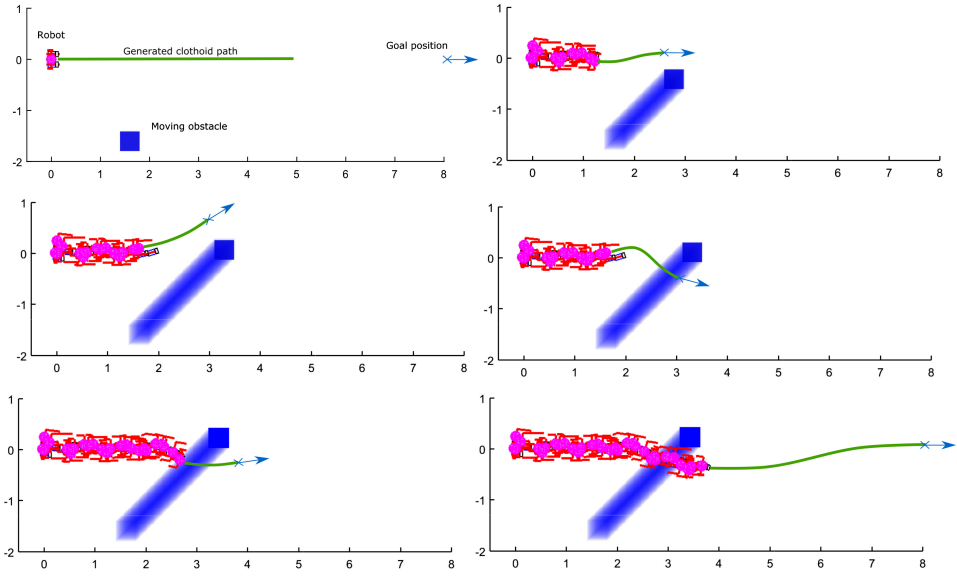


Fig. 12. Sequence of figures illustrating the online path planning for obstacle avoidance.

obstacles. The points are used to generate two distinct clothoid curves, both subject to evaluation to determine whether they provide the necessary clearance from the obstacles. The algorithm selects the clothoid curve that brings the robot closer to its final destination. This process is repeated continuously, addressing each obstacle within the robot's "visible" range. In cases where multiple potential targets exist, the algorithm opts for the one with the shortest distance from the robot's current position. Figure 12 illustrates an example of a path that updates online in case when obstacle moves and obstructs the walking direction.

From Fig. 12, we can see that the robot first updated the path to avoid it from the robot's right side, but since the obstacle continues to move, the path is updated to direct a robot to avoid the obstacle from the robot's left side.

5. Machine Learning-Based Approaches to Biped Walking

We devote this section to a contemporary approach to robot biped walking based on machine learning (ML) techniques. Initially, ML techniques were used only to optimize the parameters of the conventional feedback controllers used for biped stabilization and reference trajectory tracking. However, due to the rapid development and advances in the field of deep learning (DL), different end-to-end solutions based on ML techniques have been proposed for biped walking. Most prominent approach, which has already led to successful implementations of biped walking on real full-size humanoids, is based on reinforcement learning (RL). Therefore, we devote most of this section to present theoretical background and methodology underlying

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RL-based biped walking generation, as well as to discuss some implementation issues specific for this domain of application.

5.1. *Conventional walking controllers*

Complete pipeline for the robot bipedal locomotion includes motion planning, trajectory synthesis and motion control (in case of advanced applications usually model-based and nonlinear). Different approaches to humanoid biped walking may vary in each of the three mentioned subtasks. Regarding the motion planning, some approaches are based on footstep planners, meaning that the placement of robot feet is planned in advance according to the available environmental map and/or online perception. Other approaches provide desired dynamic stability requirements and the desired characteristics of gait at the higher level of abstraction, while the specific footstep plan emerges as a consequence.

Control system itself, may be based on different types of robust, adaptive or model-predictive controllers, aimed to provide tracking of the reference joint trajectory and usually precomputed desired ZMP trajectory. Since the usage of the full dynamics model is computationally expensive, multiple reduced-order models have been proposed, based on different conceptualizations of 2D and 3D linear and nonlinear inverted pendulums or the notion of centroidal dynamics.[29,30] However, most of these conventional approaches to humanoid biped walking share the common set of essential features. Namely, they are all model-based, thus utilizing full or reduced-order dynamic model of a robotic system and they usually integrate some form of explicitly defined dynamic balance criterion within the control framework. Based on the required type of biped walking (which may be static, quasi-dynamic or dynamic) and the selected underlying complexity of dynamic model, different stability criteria have been proposed and used in literature. This observation also stands for the two above described paradigms (i) whole-body motion approach, based on novel contact wrench condition called DBM[10] and (ii) biologically inspired approach based on concept of motion primitives[25,27] combined with ZMP criterion. Most of the conventional methods assume humanoids with flat feet, and perform reasonably well on such class of robotic systems, but the achieved motions tend to be a bit conservative and slow. Also, the constraint of constant pelvis height, induced by some reduced-complexity models of dynamics, leads to quite unnatural walking patterns.

5.2. *RL — Problem formulation*

However, due to the recent advances in DL techniques, novel paradigm for robot bipedal locomotion, primarily based on RL method achieved some promising results. The application of this paradigm for biped walking is essentially different in the sense that it is usually model-free and that it often lacks any explicitly defined stability criterion. Although, the theoretical background for this approach is not novel, since the first attempts to generate bipedal walking based on RL were performed several decades ago,[31] some practical limitations and challenges related to the transfer of the

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learned controller from simulation environment to the real hardware platform (so called sim2real gap) were only recently successfully addressed. For the most relevant introduction to RL we refer reader to Ref. 32, and for its extension to the concept of deep reinforcement learning (DRL) to Ref. 33.

Briefly speaking, RL is one of the ML paradigms, besides supervised and unsupervised learning, used by a robot which behaves in a trial-and-error manner in order to find an optimal policy $\pi(s) : \mathbf{S} \rightarrow \mathbf{A}$, i.e., optimal mapping from state space to action space, based on the acquired rewards during the multiple episodes of simulated or real experience. In other words, the policy represents a prescription for behavior, telling the robot which action should be performed in each particular state, in order to maximize the expected return, i.e., the expected cumulative reward over some desired time horizon T. Formally speaking, RL framework models the world as a discrete time Markov Decision Process, given with a tuple $(\mathbf{S}, \mathbf{A}, P, r, \gamma)$, which is a usual framework for describing sequential decision making. The world is described by a continuous state space $\mathbf{S} \in \mathbb{R}^n$ and continuous action space $\mathbf{A} \in \mathbb{R}^m$, together with a transition probability function $P : \mathbf{S} \times \mathbf{A} \times \mathbf{S} \rightarrow [0, 1]$ which defines the world dynamics in the form of probability density of transitions among different states, given that certain action has been performed. Also, there is a scalar reward function $r : \mathbf{S} \times \mathbf{A} \rightarrow \mathbf{R}$ describing the reward/punishment that an agent gets by getting to state s_{t+1} from state s_t, executing the action a. Finally, there is a discount factor $\gamma \in [0, 1)$. Discounted return, which should be maximized by optimal policy, is given as $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$.

In case of complex and high-dimensional systems, typical for robot control applications, it may be very hard to model environment's transition dynamics given with transition probabilities. In case of model-based RL approaches Monte Carlo sampling may be used to learn the transition dynamics from episodes of agent's experience and interaction with environment. However, in the context of robot biped walking it is more common to use model-free approach to RL. It is a class of RL algorithms where an agent learns to make decisions and optimizes its behavior in an environment without explicitly building a model of that environment's transition dynamics. Namely, model-free RL methods tend to be more sample-efficient and can learn directly from trial and error. Although called model-free, RL-based modeling of biped walking is usually not completely agnostic about the physics of the system, since it may significantly benefit from physical insights, mainly incorporated through the handcrafted reward function or appropriate action parametrization.

5.3. *Implementation-related issues — simulation to reality gap*

RL-based setup, related to the robot bipedal locomotion problem, has several domain-specific challenges which are briefly discussed and emphasized here. These issues are caused by the inherent nature of the physical properties of the humanoid robot like high-dimensional continuous state and action spaces, highly coupled nonlinear behavior, insufficient knowledge of true dynamic parameters (hard to

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estimate through system identification) and inability to acquire data (episodes of experience) from real robotic systems, as well by the complexity of the environment. The inability to use real robot for training is mostly related to safety problems. Namely, since RL agent learns in trial-and-error manner, similar to a child learning to walk, expensive and very sophisticated humanoid robotic systems may be severely damaged during this process. Also, the learning process is very data-hungry and requires a vast number of episodes of experience to create a representative dataset for policy learning, which would be very inefficient and time consuming to acquire on a real system. Therefore, learning of control policies is usually performed within some simulation environment, instead on the real robot.

Transfer and deployment of control policies learned in simulation environment to real hardware platforms poses multiple challenges due to significant differences in kinematic and dynamic parameters between robot models and real robots, as well as due to inherent approximate nature of modeling process unable to capture all the physical phenomena relevant for the environment description. This problem has become known in the literature as sim2real gap. Hence, first successful implementations were mostly limited to lightweight bipedal robots without upper body, like Cassie[34] or quadripedal animal-like robots.[35] Due to direct-drive actuation, low gear ratio transmission systems with higher backdrivability and lower masses, the above-mentioned differences were less influential. Speaking of humanoid robots, whose dimensions and mass distribution are comparable to those of humans, first successful transfers have only recently been reported in literature, and they have required multiple innovative techniques to be developed and applied like dynamics randomization,[36] curriculum learning,[37] action space bounding and improved modeling and estimation of actuator dynamics and latency.[38]

One of the techniques for dealing with a sim2real gap is dynamics randomization. The idea is to randomize robot dynamic and other simulation-related parameters such as mass, inertia properties, friction and damping coefficients, to improve robustness of the policy. Also, different types of noises are introduced into simulation to model imperfect sensor measurements in real-world data, noise and latency in actuator control signals and environmental noise to represent external disturbances and uncertainties related to the terrain. Training in the randomized simulation environment increases the success rate of the policy transfer to the real world.

5.4. *Action parametrization and reward function design*

Regarding the control architecture, most often RL-based motion controllers are hierarchical, consisting of high-level lower-frequency RL controller which maps the robot state vector into some parametrized action output (usually reference joint position vector or relative joint position offsets), and low-level higher-frequency local feedback controller which maps the output from RL controller into driving joint torques. However, different action parametrizations which define the output from RL controller (and corresponding types of feedback controllers) have been investigated

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in literature, including (i) reference joint positions, (ii) reference joint velocities, (iii) muscle activations and (iv) joint torques.[39] The results suggested that first three action parametrizations which include local feedback controllers (PD, D and Muscle Tendon Units) have advantage over the last direct force control which directly outputs joint torques. Due to its simplicity and effectiveness, proportional-derivative (PD) controller has been widely adopted as an appropriate choice in the research community.

Insufficiently accurate actuator torque tracking, most probably due to the induced back electromotive force (back EMF), has been identified as one of the major causes of the sim2real gap, even more dominant then inaccurate modeling of mass distribution.[40] In the same paper a solution has been proposed, based on a cascade PD–PI controller with outer position loop and inner motor current loop. PD controller maps outputs from learned RL control policy (desired joint positions) into desired torques, while the inner PI controller modifies these desired torques based on the motor current feedback and provides control signals for actuators. Additionally, back EMF was simulated as a parametrized damping term, degrading the calculated reference torque (from PD controller), and used for policy learning during training phase.

RL-based model of robot biped walking includes both continuous state space (e.g., joint positions, joint velocities, etc.) and action space (joint torques or forces). One approach to dealing with continuous space problems is to discretize continuous variables, which in case of high-dimensional spaces leads to exponential combinatorial explosion known as curse of dimensionality. Representing the control policy in continuous action spaces may be challenging and it is usually based on usage of universal function approximators, like deep neural networks. A class of algorithms for model-free RL, particularly suitable for continuous action spaces is the class of policy gradient methods. These methods apply direct policy parametrization using neural networks, supporting both deterministic policies which output the particular action $\pi(s) : \mathbf{S} \rightarrow \mathbf{A}$, and stochastic policies which output probability distributions over possible actions $\pi(s, a) : \mathbf{S} \times \mathbf{A} \rightarrow [0, 1]$. Policy gradient methods, in their basic form, directly optimize a parametrized control policy by gradient ascent technique. More precisely, particular algorithms within this class differ according to the goal of their learning process. Some of them perform direct optimization of policy parameters, usually in on-policy fashion, while others perform Q-learning in an off-policy manner, i.e., they learn an approximation of the optimal action-value function, which indirectly optimizes the corresponding policy.[41] The second approach is less stable, but more sample-efficient. There are some hybrid approaches also, which combine these two learning goals. Some state-of-the-art gradient-based algorithms which learn stochastic policies include Trust Region Policy Optimization (TRPO)[42] and Proximal Policy Optimization (PPO).[43]

Furthermore, there is a special subgroup of policy gradient methods, called actor-critic methods,[44] which introduce two separate function approximators (usually deep neural networks). The critic approximator estimates the value function (action-value

Q or state value V), while the actor approximator updates the control policy distribution, guided by the critic's suggestions (in a gradient-like manner). Two state-of-the-art approaches, based on this idea, are Deep Deterministic Policy Gradient (DDPG)[45] and Soft Actor-Critic (SAC).[46] These two algorithms perform both policy optimization and Q-learning in parallel. The first algorithm outputs the deterministic target policy, and the second stochastic one. Policy gradient methods are very popular in robotics, for applications like manipulation control, path planning and biped walking.

Designing an appropriate reward function is essential part of RL approach, since the reward should encourage the robot to select actions that would lead to stable walking. It should also provide an appropriate balance between exploration and exploitation of action space. Although it is possible to include some explicitly defined balance criterion, like ZMP, in the reward function, it is more common to handcraft the reward function based on the more descriptive and indirect balance indicators like forward progress, staying upright, energy efficiency and avoiding limits, collisions or falls. In the first case, rewards could be directly analytically defined for keeping the ZMP within certain desired bounds or even for following desired ZMP trajectories. However, for most of the RL-based approaches to robot bipedal walking there is a specific absence of explicit criterion of dynamic balance preservation within the learning framework.

In the literature, we can identify two main approaches based on DRL. The first approach uses Motion Capture (MoCap) systems to provide rich and dense reference motion data by recording the human subject performing the relevant task, and thus providing an expert demonstration. This captured reference motion is then used to define the reward function for the RL process, in order to learn a policy which generates robot motion as similar as possible to the recorded one. In Ref. 47, a composite two terms reward function has been proposed, consisting of (i) an imitation objective (which encourages the robot to match the reference motion style) and (ii) task objective (which tends to fulfill task-specific behaviors). MoCap-based approach imposes the problem of kinematic mapping between generally different kinematic structures of human demonstrator and corresponding humanoid robot, as well as motion interpolations. Second approach does not use reference trajectory data at all, but exploits some properties of the human gait, like symmetry and periodicity, to handcraft the suitable reward function in order to provide dynamically stable and naturally looking gait, including the above-mentioned indirect balance indicators.[34,38,48]

5.5. *State-of-the-art RL-based walking controllers*

In the rest of this section, we will discuss several state-of-the-art DRL-based approaches to robot biped walking and balancing, with particular emphasis (but not exclusively) on the research which included successful control policy transfer to the real humanoid robot.

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In Ref. 38, DRL-based approach to omnidirectional biped locomotion of humanoid robots has been proposed. Unlike the most of former approaches which assumed multiple, independently trained control policies, this approach proposes a single control policy for realization of different modes of biped walking, like forward or backward walking and turning. Robot operation in a human-centered and dynamic unstructured environment requires the ability of walking on both flat and uneven terrain, slopes, stairs and curved paths. It also required forward and backward walking, as well as the ability to stand still, keeping the balance. Smooth transition among different modes is crucial since the switching between different controllers, trained for different modes, may induce instability in the system. This problem is addressed by using a single policy for all walking modes. The proposed approach eliminated the need for reference MoCap reference data by introducing the nominal (standing) pose and applying the curriculum learning method through gradual incremental increase of task complexity, implemented as a so called velocity scheduler. Reward function consisted of four terms representing: (i) desired velocity tracking, (ii) pose regularization (deviation from nominal standing pose), (iii) maintaining the minimal base height and (iv) providing minimal foot clearance. Sim2real problem was addressed combining multiple techniques like system identification, noise injection, dynamics randomization, modeling of actuation latency and output filtering. Actor-critic architecture was used and stochastic control policy was trained using the combination of PPO and Generalized Advantage Estimator.[38] The proposed approach was validated on real hardware, NimbRo-OP2X robot with 18 Degrees Of Freedom (DOF), performing omnidirectional walk in presence of external disturbances.

The authors in Refs. 48 and 39 present DRL approach to biped walking of human-sized humanoid robot, utilizing a single control policy to achieve different modes of omnidirectional walking both on flat and uneven terrains. In Ref. 48, the proposed DRL framework was validated in realistic simulation environment on models of HRP-5P and JVRC-1 robots. Later in Ref. 39, the framework was extended with additional feedback controller to overcome the difficulties related to sim2real gap, and successful implementation of end-to-end policy on real current-controlled humanoid was reported. Namely, in Ref. 48, a two-stage hierarchical controller has been proposed, consisting of a higher-level DRL controller, which provides reference joint positions, and a lower-level PD controller which generates driving torques. An external footstep planner was used to provide input information to the DRL controller, in terms of the desired footprints and a corresponding base heading (orientation) for the two future steps. So, the proposed approach does not apply blind proprioceptive-based locomotion, but rather exploits the available terrain data to plan the reference footsteps in advance. However, this approach does not require reference motion data acquired through human demonstrations. Instead, it uses engineered multi-term reward function which incorporates the idea of periodic reward to achieve and encourage the symmetrical and periodic nature of biped gait. Reward function also contains several additional terms which encourage the robot to: (i) progress along the

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planned footsteps as precise as possible, (ii) maintain the reference base height and (iii) maintain the upright posture. It also contains some penalties to avoid abrupt changes in driving torques and predicted actions, as a sort of regularization. The proposed DRL framework is based on actor-critic method with stochastic parametrized policy. Both the actor and critic networks are represented with Multi-Layer Perceptron architectures. Control policy was trained using PPO algorithm. Curriculum learning method was applied to learn the control policy for stairs climbing, by gradually increasing the complexity of the terrain. In Ref. 39, inappropriate motor torque tracking, due to high gear ratio, was identified as a dominant source of sim2real gap. To address the issue additional current-based feedback controller was introduced into the control architecture and simulated motor back EMF was applied as torque feedback during training, i.e., policy learning. The approach was validated on real HRP-5P humanoid.

In Ref. 49, authors investigate the possibilities of using DRL framework for humanoid balancing strategies. It presents hierarchical control framework combining high-level controller based on DRL (particularly DDPG algorithm), which learns an optimal control policy to produce reference joint angles, with low-level proportional derivative (PD) controller used to generate driving joint torques. The proposed controller provides emergent human-like balancing behaviors, including knee-locks, active ankle push offs and heel/toe tipping. This learning-based framework addresses the well-known problem of conventional ZMP-based approaches which are very limited in case of foot tilting, when ZMP reaches the boundary of the support polygon and the new underactuated DOF is introduced in the system, since no torque can be directly applied around pivoting axes. Hence, these conventional approaches are not able to exploit the full potentials of foot tilting behaviors, like humans do. Reward function of the RL process was motivated by physical insights based on ideas of capture point theory and LIPM.

The authors in Ref. 34 present model-free RL framework for generation of various walking behaviors of bipedal robot Cassie, including forward and lateral walking and turning. The proposed approach uses reference motion, represented with a gait library of diverse parametrized motions generated using Hybrid Zero Dynamics (HZD). Previously, the reference motions have been provided using conventional model-based walking controllers.[50] Control policies were then trained to provide corrections of these reference motions, the so-called residual control.[51] The problem with this approach is that the performance of the control policy is limited in the sense that it cannot overcome and extend the capabilities of the teacher, i.e., the traditional reference controller which is usually too conservative. Introduction of HZD-based reference motion increases the robustness and versatility of behaviors that the robot can learn. In Ref. 34, hierarchical control architecture has been proposed combining high-level RL controller and low-level PD controller which generates driving torques. Control policy has been trained using PPO algorithm and it provides reference joint positions as output. Composite reward function encourages the robot to minimize the motor position error, track reference pelvis motion, reduce energy

consumption and minimize vertical contact forces. Sim2real gap was addressed combining system identification and domain randomization, including randomization of robot and environment parameters, sensor noise and control signal latency.

5.6. *Future perspectives*

The above discussed papers evidence very dynamic and accelerating development in the field of learning-based robot locomotion. This approach already enabled very agile and versatile natural looking motions, like omnidirectional walk on flat and uneven terrain, turning, balancing and goal-oriented jumping. We believe that there is a huge potential in hybrid methodological approach which would fully exploit our knowledge of robot physics to guide the development of ML-based algorithms, instead of a pure data-driven black-box approach. Also, the learning component extends and overcomes our inherent limitations as designers to capture all the complexity of system dynamics and the surrounding environments in analytical form, by discovering latent patterns in different types of reference data, like human demonstrations. We anticipate that in the future, cognitively inspired models of action-perception coupling, human decision making, memory and attentional mechanisms, as well as understanding of situational context will play a more prominent role and will be tightly integrated with motion generating models, in a unified computational framework.

6. Conclusion

Development of humanoids in last decades of previous and beginning of this century was very intensive. Expectations were very high, achievements much more modest. Humanoid motion, especially (bipedal) walking, is one of the basic aspects of its activity. This becomes especially significant when we know that there was (and still is) an intention to integrate humanoids into the human living and working environment. In this paper, we aimed to summarize last 20 years from standpoint of humanoid biped motion and to present how far we have gone, what the remaining open challenges are and what we can expect in the further development.

Basic requirement for humanoids in future is motion in unstructured environment. This requires development of complex sensory and cognitive capabilities to support and enable successful robot localization in its environment, interpretation and understanding of its situational context and goal-oriented planning and decision making. The last item is directly related to action. We would like to emphasize that, due to inherent dynamic nature of human-centered environment, all the decisions and corresponding actions have to be generated online, thus making offline pre-programming inadequate. The paper gives an overview of the attempts to solve this task, and also gives a detailed description of the approach based on composition of motion from basic movements — primitives (for example, leg bending, extension, etc.).

Another motion-related topic is motion in the presence of multiple spatially distributed contacts. Humans often use additional contacts to facilitate motion: lean

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against the wall, hold on to the handrail, so it can be expected that it would also be useful for humanoids in similar situations. Considering that the ZMP is an indicator of dynamic balance during walking (the robot's ability to constantly maintain consistently its vertical body position, i.e., to be "on its feet"), there was a tendency to extend the concept of ZMP to cases where the contacts are spatially dispersed. This paper gives an overview of the basic attempts to generalize the concept of ZMP, and the approach based on computing 5D convex hull is presented in detail.

The last section is about contemporary approaches to robot bipedal locomotion, based on ML. Although attempts to imitate the process by which a baby learns to walk have existed for a long time, the first results have only recently been achieved. RL seems to be the most promising method. Most of the learning is performed in simulation and then transferred to real robot. However, some discrepancies appear due to differences between a real robot and simulated one. This is known as "sim2real gap", which is still an active field of research.

When we think only of motion, we think that in the future we can expect a significantly higher participation of learning in the training of robots, in both: in the fine-tuning of movement (the robot needs to learn how to move in a human-like manner) and in training to perform work tasks. In order to achieve such a thing, our opinion is that we need significantly improved learning algorithms and control systems.

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References

1. M. Vukobratović and B. Borovac, Zero-moment point — Thirty five years of its life, *Int. J. Humanoid Robot.* **1**(1) (2004) 157–173.
2. M. Vukobratović, B. Borovac and D. Šurdilović, Zero moment point — Proper interpretation and new applications, in *Proc. 2nd IEEE-RAS Int. Conf. Humanoid Robots* (IEEE, 2001), pp. 237–244.
3. M. Vukobratović, B. Borovac and V. Potkonjak, ZMP: A review of some basic misunderstandings, *Int. J. Humanoid Robot.* **3**(2) (2006) 153–175.
4. K. Harada, S. Kajita, K. Kaneko and H. Hirukawa, Dynamics and balance of a humanoid robot during manipulation tasks, *IEEE Trans. Robot.* **22**(3) (2006) 568–575.

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5. H. Hirukawa, S. Hattori, K. Harada, S. Kajita, K. Kaneko, F. Kanehiro, K. Fujiwara and M. Morisawa, A universal stability criterion of the foot contact of legged robots - Adios ZMP, in *Proc. 2006 IEEE Int. Conf. Robotics and Automation* (IEEE, 2006), pp. 1976–1983.
6. P. Sardain and G. Bessonnet, Forces acting on a biped robot. Center of pressure-zero moment point, *IEEE Trans. Syst. Man Cybern. A, Syst. Humans* **34**(5) (2004) 630–637.
7. S. Caron, Q. C. Pham and Y. Nakamura, ZMP support areas for multicontact mobility under frictional constraints, *IEEE Trans. Robot.* **33**(1) (2017) 568–575.
8. T. Brecelj and T. Petrič, Zero moment line — Universal stability parameter for multi-contact systems in three dimensions, *Sensors* **22**(15) (2022) 5656.
9. P. Ferrari, L. Rossini, F. Ruscelli, A. Laurenzi, G. Oriolo, N. Tsagarakis and E. Mingo, Multi-contact planning and control for humanoid robots: Design and validation of a complete framework, *Robot. Auton. Syst.* **166** (2023) 104448.
10. M. Nikolić, B. Borovac and M. Raković, Dynamic balance preservation and prevention of sliding for humanoid robots in the presence of multiple spatial contacts, *Multibody Syst. Dyn.* **42** (2018) 197–218.
11. S. Caron, Q. C. Pham and Y. Nakamura, Stability of surface contacts for humanoid robots: Closed-form formulae of the contact wrench for rectangular support areas, in *IEEE Int. Conf. Robotics and Automation* (IEEE, 2015), pp. 5107–5112.
12. J. Farkas, Theorie der einfachen ungleichungen, *J. Reine Angew. Math. (Crelles J.)* **124** (1902) 1–27.
13. M. Nikolić, B. Borovac, M. Raković and S. Savić, A further generalization of task-oriented control through tasks prioritization, *Int. J. Humanoid Robot.* **10**(3) (2013) 1350012.
14. M. Vukobratović and D. Juricic, Mathematical modeling of a bipedal walking system, in *Proc. ASME Winter Annual Meeting* (ASME, 1972), pp. 26–30.
15. S. Kajita, F. Kanehiro, K. Kaneko, K. Fujiwara, K. Harada, K. Yokoi and H. Hirukawa, Biped walking pattern generation by using preview control of zero-moment point, in *2003 IEEE Int. Conf. Robotics and Automation* (IEEE, 2003), pp. 1620–1626.
16. J. E. Pratt and S. V. Drakunov, Derivation and application of a conserved orbital energy for the inverted pendulum bipedal walking model, in *Proc. 2007 IEEE Int. Conf. Robotics and Automation* (IEEE, 2007), pp. 4653–4660.
17. J. H. Park and K. D. Kim, Biped robot walking using gravity-compensated inverted pendulum mode and computed torque control, in *Proc. 1998 IEEE Int. Conf. Robotics and Automation* (IEEE, 1998), pp. 3528–3533.
18. R. C. Luo, J. Sheng, C. C. Chen, P. H. Chang and C. I. Lin, Biped robot push and recovery using flywheel model based walking perturbation counteraction, in *2013 13th IEEE-RAS Int. Conf. Humanoid Robots (Humanoids)* (IEEE, 2013), pp. 50–55.
19. S. Wang, S. Piao, X. Leng and Z. He, Learning 3D bipedal walking with planned footsteps and Fourier series periodic gait planning, *Sensors* **23**(4) (2023) 1873.
20. F. M. Smaldone, N. Scianca, L. Lanari and G. Oriolo, From walking to running: 3D humanoid gait generation via MPC, *Front. Robot. AI* **9** (2022) 876613.
21. K. Mombaur, Using optimization to create self-stable human-like running, *Robotica* **27** (2009) 321–330.
22. S. Feng, E. Whitman, X. Xinjilefu and C. G. Atkeson, Optimization based full body control for the atlas robot, in *2014 IEEE-RAS Int. Conf. Humanoid Robots* (IEEE, 2014), pp. 120–127.
23. T. Keenan and S. Evans, *An Introduction to Child Development* (SAGE Publications, 2009).
24. K. E. Adolph, B. Vereijken and P. E. Shrout, What changes in infant walking and why, *Child Dev.* **74**(2) (2003) 475–497.

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25. M. Raković, B. Borovac, M. Nikolić and S. Savić, Realization of biped walking in unstructured environment using motion primitives, *IEEE Trans. Robot.* **30**(6) (2014) 1318–1332.
26. M. Raković, B. Borovac, S. Savić and M. Nikolić, Parameters adaptation of motion primitives for achieving more efficient humanoid walk, in *2014 IEEE 12th Int. Symp. Intelligent Systems and Informatics (SISY)* (IEEE, 2014), pp. 233–237.
27. M. Raković, S. Savić, J. Santos-Victor, M. Nikolić and B. Borovac, Human-inspired online path planning and biped walking realization in unknown environment, *Front. Neurorobot.* **13** (2019) 36.
28. G. Arechavaleta, J.-P. Laumond, H. Hicheur and A. Berthoz, An optimality principle governing human walking, *IEEE Trans. Robot.* **24** (2008) 5–14, doi: 10.1109/TRO.2008.915449.
29. J. Reher and A. D. Ames, Dynamic walking: Toward agile and efficient bipedal robots, *Annu. Rev. Control Robot. Auton. Syst.* **4** (2021) 535–572.
30. S. Kajita, H. Hirukawa, K. Harada and Y. Yokoi, Biped walking, in *Introduction to Humanoid Robotics*, Springer Tracts in Advanced Robotics, Vol. 101 (Springer, Berlin, 2014), pp. 105–158.
31. H. Benbrahim and J. A. Franklin, Biped dynamic walking using reinforcement learning, *Robot. Auton. Syst.* **22** (1997) 283–302.
32. R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd edn. (The MIT Press, 2018).
33. A. Plaat, *Deep Reinforcement Learning*, 1st edn. (Springer, Singapore, 2022).
34. Z. Li, X. Cheng, X. B. Peng, P. Abbeel, S. Levine, G. Berseth and K. Sreenath, Reinforcement learning for robust parameterized locomotion control of bipedal robots, in *2021 IEEE Int. Conf. Robotics and Automation (ICRA)* (IEEE, 2021), pp. 2811–2817.
35. M. Aractingi, P.-A. Léziart, T. Flayols, J. Perez, T. Silander and P. Souères, Controlling the Solo12 quadruped robot with deep reinforcement learning, *Sci. Rep.* **13** (2023) 11945.
36. F. Muratore, F. Ramos, G. Turk, W. Yu, M. Gienger and J. Peters, Robot learning from randomized simulations: A review, *Front. Robot. AI* **9** (2022) 799893.
37. S. Narvekar, B. Peng, M. Leonetti, J. Sinapov, M. E. Taylor and P. Stone, Curriculum learning for reinforcement learning domains: A framework and survey, *J. Mach. Learn. Res.* **2**(1) (2020) 7382–7431.
38. D. Rodriguez and S. Behnke, DeepWalk: Omnidirectional bipedal gait by deep reinforcement learning, in *2021 IEEE Int. Conf. Robotics and Automation (ICRA)* (IEEE, 2021), pp. 3033–3039.
39. R. P. Singh, Z. Xie, P. Gergondet and F. Kanehiro, Learning bipedal walking for humanoids with current feedback, preprint (2023), arXiv:2303.03724v2.
40. X. B. Peng and M. van de Panne, Learning locomotion skills using DeepRL: Does the choice of action space matter? in *Proc. ACM SIGGRAPH/Eurographics Symp. Computer Animation (SCA '17)* (Association for Computing Machinery, New York, 2017), pp. 1–13.
41. Part 2: Kinds of RL Algorithms — Spinning up documentation (2023), https://spinningup.openai.com/en/latest/spinningup/rl_intro2.html.
42. J. Schulman, S. Levine, P. Abbeel, M. Jordan and P. Moritz, Trust region policy optimization, in *Proc. 32nd Int. Conf. Machine Learning* (PMLR, 2015), pp. 1889–1897.
43. J. Schulman, F. Wolski, P. Dhariwal, A. Radford and O. Klimov, Proximal policy optimization algorithms, preprint (2017), arXiv:1707.06347.
44. V. R. Konda and J. N. Tsitsiklis, Actor-critic algorithms, in *Proc. 12th Int. Conf. Advances in Neural Information Processing Systems* (MIT Press, Cambridge, 1999), pp. 1008–1014.

45. D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra and M. Riedmiller, Deterministic policy gradient algorithms, in *Proc. 31st Int. Conf. Machine Learning (ICML'14)* (PMLR, 2014), pp. 387–395.
46. T. Haarnoja, A. Zhou, P. Abbeel and S. Levine, Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor, in *Proc. 35th Int. Conf. Machine Learning (ICML 2018)* (PMLR, 2018), pp. 1861–1870.
47. X. B. Peng, P. Abbeel, S. Levine and M. van de Panne, DeepMimic: Example-guided deep reinforcement learning of physics-based character skills, *ACM Trans. Graph.* **37**(4) (2018) 1–14.
48. R. P. Singh, M. Benallegue, M. Morisawa, R. Cisneros and F. Kanehiro, Learning bipedal walking on planned footsteps for humanoid robots, in *2022 IEEE-RAS 21st Int. Conf. Humanoid Robots (Humanoids)* (IEEE, 2022), pp. 686–693.
49. C. Yang, T. Komura and Z. Li, Emergence of human-comparable balancing behaviours by deep reinforcement learning, in *2017 IEEE-RAS 17th Int. Conf. Humanoid Robotics (Humanoids)* (IEEE, 2017), pp. 372–377.
50. Z. Xie, G. Berseth, P. Clary, J. Hurst and M. van de Panne, Feedback control for Cassie with deep reinforcement learning, in *2018 IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)* (IEEE, 2018), pp. 1241–1246.
51. T. Johannink, S. Bahl, A. Nair, J. Luo, A. Kumar, M. Loskyll, J. A. Ojea, E. Solowjow and S. Levine, Residual reinforcement learning for robot control, in *2019 Int. Conf. Robotics and Automation (ICRA)* (IEEE, 2019), pp. 6023–6029.



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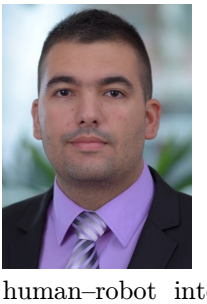
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